

FOUNDATIONS OF A THEORY OF COMPOSABLE ABSTRACTIONS (COMPANION TO THE STATIONARY-BARRIER PROGRAMME)

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ABSTRACT. We are missing a theory that answers *which structures let knowledge transfer*: a counterpart to computability (what can be computed), complexity (at what cost), and learnability (from how many samples). This companion lays the foundations of one. We fix the central object—an *abstraction* $(\mathcal{Z}, \phi)$ and its *defect* $\delta(\phi, \mathcal{F}) = \sup_{f \in \mathcal{F}} \inf_g \|f - g \circ \phi\|$ relative to a concept class $\mathcal{F}$ —and prove the structural facts on which a full theory must rest. The load-bearing observation is that the defect depends on ϕ *only through the closed subspace* $V_\phi = L^2(\sigma(\phi))$, with the optimal decoder given by the conditional expectation $\mathbb{E}[f \mid \phi]$ (theorem 2.1). From it, three of the four open problems reduce to operator-theoretic and geometric statements that can be settled or sharply framed: **(i) Compression.** The compression–defect frontier is *exactly* a nonlinear width of $\mathcal{F}$; it is non-degenerate only under a decoder budget, and its decay rate is the effective dimension d_{loc}^* of the parent programme (theorems 3.1 and 3.2). **(ii) Reusability.** The reusability axiom, false as literally stated, is true *precisely on the cone* V_ϕ , with an irreducible bias $\|(I - P_\phi)f'\|$ outside it (theorem 4.3). **(iii) Transfer.** The right metric on concept classes is the subspace gap between their width-optimal subspaces, and transfer is bounded by it: $\delta(\phi_{\mathcal{F}_1}, \mathcal{F}_2) \leq d_n(\mathcal{F}_2) + R\Theta(V_1, V_2)$ (theorem 5.2)—a theorem of exactly the form the open problem requested. **(iv) Composability.** Naively stacking abstractions against a *fixed* class can only raise the defect (theorem 6.1); the resolution is a filtration of level-tasks, under which composition reduces to the parent paper’s compositional exponent $d_{\text{loc}}^* = \max_i d_i$ (theorem 6.4). The *necessity* direction of the existence conjecture is proved for the Besov class $\mathcal{F}_{s,p}^{\text{Bsv}}$ with $p < 2$: via an explicit wavelet counterexample the recomposition constant of any *fixed-subspace* serial chain collapses as $c_{\text{recomp}} \leq \varepsilon^{\sigma/s} \rightarrow 0$ ($\sigma = 1/p - 1/2$), showing that fast nonlinear width does not imply serial factorisability (theorems 6.10 and 6.11); the sharper rate $\varepsilon^{2\sigma}$ for budget-constrained adaptive chains is stated only as a conjecture (theorem 6.13), and fully adaptive chains do not collapse at all (theorem 6.12). We set up the *learnability index* $\kappa(\mathcal{F})$ (section 7) and then *discharge* its central hypothesis for a named class: for the Gaussian multi-index model the interaction gap equals the smallest singular value of the link’s expected Hessian, $\Delta_2 = \sigma_{\min}(\mathbb{E}[\nabla^2 g])$ (theorem 8.3), so link ellipticity forces $\Delta \geq \rho$ over the whole class (theorem 8.4) and its closure reproduces the statistical-query barrier (theorem 8.5)—showing concretely that the learnability gap, previously required as a condition, follows from the geometry of the link function for this class, and via a stability lemma plus a second-order-score argument extending learnability up the chain unconditionally on score-regular intermediate measures (theorems 8.7 and 9.3). We close by *proving* the local-to-global *characterization* problem (P1): defining the automorphism sheaf $\underline{\text{Aut}} \mathcal{S}$ as the gauge group of L^2 -isometric representation changes, establishing the Čech coboundary calculus $\delta^1 g_{abc} = g_{bc} \circ g_{ab} \circ g_{ca}$, and proving the main bound $\delta_{\text{global}} \leq \varepsilon_{\text{local}} + C(D, n) \cdot R(\mathcal{F}) \cdot \|[g]\|_{\tilde{H}^1}$ via a triangle inequality for subspace gaps along the nerve of the cover (theorem 12.7). The characterization invariant is $I(\phi, \mathcal{F}) = ([P_\phi], [g])$, and goodness is equivalent to local width achievement plus $[g] = [\text{id}]$ in $\tilde{H}^1$ (theorem 12.9). The P2–P1 bridge closes through $c_{\text{recomp}} = \sqrt{1 - \|[g]\|_{\tilde{H}^1}^2}$ (theorem 12.11), so all four problems P1–P4 are now connected by the single quantity $\|[g]\|_{\tilde{H}^1}$. We keep the parent programme’s grading discipline: every statement is marked theorem, conjecture, or programme according to what its proof earns.

1. INTRODUCTION: THE MISSING FOURTH THEORY

Three theories of information processing are mature. Computability theory answers *what* can be computed; complexity theory, *how expensively*; statistical learning theory (PAC, minimax), *from how many samples*. A fourth question has no comparable theory: *which structures let knowledge be reused*—what makes a representation of data good not for one task but for a family, and what makes such a representation composable into a hierarchy. Empirically this is the entire content of representation learning and transfer; theoretically it is a collection of special cases without a unifying object.

This companion proposes the object and proves the foundational facts. The immediate motivation came from the phase-transition analysis in the companion scaling paper: once it became clear that the capacity threshold N_c is finite only when the target has a compositional flag, the next question was unavoidable—what, precisely, is the object that a flag represents, and how does it transfer across task families? Trying to answer that question with standard kernel or PAC tools kept running into the same wall: those tools describe performance on one task, not reuse across a class. The defect is the missing quantity.

The setting is deliberately minimal. Let $\mathcal{X} \subset \mathbb{R}^d$ carry a probability measure μ , and let $\mathcal{F} \subset L^2(\mathcal{X}, \mu)$ be a *concept class*: the family of tasks we care about.

Definition 1.1 (Abstraction and defect). An *abstraction* is a pair $(\mathcal{Z}, \phi)$ with a representing space $\mathcal{Z}$ and a measurable *feature map* $\phi : \mathcal{X} \rightarrow \mathcal{Z}$. Its *defect* relative to $\mathcal{F}$, under a class of admissible *decoders* $\mathcal{G} \subseteq \{g : \mathcal{Z} \rightarrow \mathbb{R}\}$, is

$$(1) \quad \delta_{\mathcal{G}}(\phi, \mathcal{F}) := \sup_{f \in \mathcal{F}} \inf_{g \in \mathcal{G}} \|f - g \circ \phi\|_{L^2(\mu)}.$$

When $\mathcal{G} = L^2(\mathcal{Z}, \phi_{\#}\mu)$ (all square-integrable decoders) we write $\delta(\phi, \mathcal{F})$ and call it the *unconstrained defect*.

The defect is the worst-case error on the hardest task in $\mathcal{F}$, under the best decoder. A good abstraction wants δ small *and* $\mathcal{Z}$ small; but smallness of the defect alone is vacuous (take $\phi = \text{id}$, $\delta = 0$, useless). The parent note isolates three further requirements.

- A1 Compression.** The effective dimension of $\mathcal{Z}$ is $\ll d$. We formalise this through the parent programme's d_{loc}^* .
- A2 Reusability.** For tasks f' in an enlarged family $\mathcal{F}' \supseteq \mathcal{F}$, learning from $\mathcal{Z}$ is no slower than learning from $\mathcal{X}$.
- A3 Composability.** Abstractions stack, $\mathcal{X} \xrightarrow{\phi_1} \mathcal{Z}_1 \xrightarrow{\phi_2} \dots \xrightarrow{\phi_L} \mathcal{Z}_L$, with ϕ_{i+1} seeing *only* $\mathcal{Z}_i$, the defect controlled at each level, and the composite efficiently learnable.

Four problems organise the theory. **(P1) Characterization:** a computable invariant $I(\phi, \mathcal{F})$ that is necessary and sufficient for ϕ to be a good abstraction. **(P2) Existence:** when does a depth- L composable abstraction of small defect exist? **(P3) Learnability:** from i.i.d. samples of a *single* $f \in \mathcal{F}$, when can one *build* a ϕ satisfying A1–A3, and at what sample cost? **(P4) Transfer:** a metric $\text{dist}(\mathcal{F}_1, \mathcal{F}_2)$ on concept classes with $\delta(\phi_{\mathcal{F}_1}, \mathcal{F}_2) \leq C \text{dist}(\mathcal{F}_1, \mathcal{F}_2)$.

What this paper does. We do not solve P1–P4; we lay the foundation that makes them precise and settles the parts that are settleable. The organising device is a single identity (section 2): the defect sees ϕ only through the closed subspace it generates, and the optimal decoder is a conditional expectation. From it: section 3 pins the compression–defect frontier to a nonlinear width and recovers d_{loc}^* ; section 4 repairs and proves A2; section 5 proves a transfer bound of exactly the form P4 asks; section 6 proves a structural obstruction to naive composition and reframes A3; section 7 sets up the learnability index for P3 and ties it to the parent barrier; section 12 frames P1 as a cohomological gluing obstruction. Section 13 is the status table and roadmap.

Relation to the parent programme. The parent paper [1] studies, in the language above, the single concept class $\mathcal{F} = \mathcal{C}_{d_{\text{loc}}^*}^s$ of compositional Sobolev targets, and its results—the minimax barrier, the effective dimension d_{loc}^* , the compositional exponent $d_{\text{loc}}^* = \max_i d_i$, the interaction gap Δ , the statistical-query hardness at $\Delta = 0$ —are the first bricks in P3 for that class. We treat its d_{loc}^* as the *rate* of an abstraction’s compression frontier (theorem 3.2), its Δ as the order parameter of the learnability index (section 7), and its compositional theorem as the existence half of A3 (section 6). Where the parent paper proves things about one class, here we ask what is true about the *operation of abstraction itself*, for any class.

Grading discipline. Theorems and propositions are proved under stated hypotheses (mostly they are elementary functional analysis, which is the point: the right object makes them elementary). Conjectures are precise but unproved. Programmes are research directions with a stated first step. Section 13 collects the grades.

2. FOUNDATIONS: THE DEFECT IS A PROJECTION

Everything rests on the following observation, which we state carefully because the distinction between constrained and unconstrained decoders marks precisely where the theory either becomes trivial or becomes interesting.

Theorem 2.1 (The defect is a projection residual). *Fix a measurable $\phi : \mathcal{X} \rightarrow \mathcal{Z}$ and let $V_\phi := L^2(\mathcal{X}, \sigma(\phi), \mu)$ be the closed subspace of $L^2(\mu)$ consisting of (μ -equivalence classes of) $\sigma(\phi)$ -measurable functions, with orthogonal projection P_ϕ . Then for every $f \in L^2(\mu)$,*

$$(2) \quad \inf_{g \in L^2(\mathcal{Z}, \phi_{\#}\mu)} \|f - g \circ \phi\|_{L^2(\mu)}^2 = \|f - P_\phi f\|^2 = \mathbb{E}_\mu[\text{Var}(f \mid \phi)],$$

and the infimum is attained at the conditional expectation $g^ \circ \phi = \mathbb{E}[f \mid \phi]$ (the Doob–Dynkin factorisation guarantees g^* measurable). Consequently the unconstrained defect*

$$(3) \quad \delta(\phi, \mathcal{F})^2 = \sup_{f \in \mathcal{F}} \|(I - P_\phi)f\|^2$$

depends on ϕ only through the subspace V_ϕ (equivalently, only through the σ -algebra $\sigma(\phi)$).

Proof. The set $\{g \circ \phi : g \in L^2(\mathcal{Z}, \phi_{\#}\mu)\}$ equals V_ϕ : a function is a.s. equal to a measurable function of ϕ iff it is $\sigma(\phi)$ -measurable (Doob–Dynkin), and the change of variables $\|g \circ \phi\|_{L^2(\mu)} = \|g\|_{L^2(\phi_{\#}\mu)}$ makes the map $g \mapsto g \circ \phi$ an isometry onto V_ϕ , whose image is closed. Minimising $\|f - v\|$ over a closed subspace V_ϕ of a Hilbert space is solved by the orthogonal projection $v = P_\phi f$, and orthogonal projection onto $L^2(\sigma(\phi))$ is conditional expectation $\mathbb{E}[\cdot \mid \sigma(\phi)]$; the residual norm squared is $\mathbb{E}[(f - \mathbb{E}[f \mid \phi])^2] = \mathbb{E}[\text{Var}(f \mid \phi)]$. As P_ϕ is determined by V_ϕ , so is the supremum in (3). $\square$

Definition 2.2 (Equivalent abstractions). Two abstractions $(\mathcal{Z}, \phi)$ and $(\mathcal{Z}', \phi')$ are *equivalent*, written $\phi \sim \phi'$, if $V_\phi = V_{\phi'}$, i.e. $\sigma(\phi) = \sigma(\phi')$ up to μ -null sets. The *invariant* of an abstraction is its equivalence class $[\phi]$, represented by the projection P_ϕ .

Remark 2.3 (The natural category of abstractions). Theorem 2.2 is the categorical content the parent note anticipated under “natural transformations as equivalent abstractions.” Feature maps over $\mathcal{X}$ form a category $\text{Abs}(\mathcal{X})$ whose objects are measurable maps and whose morphisms $\phi \rightarrow \psi$ are measurable h with $\psi = h \circ \phi$ (a factorisation: ψ is a *coarsening* of ϕ). A morphism $\phi \rightarrow \psi$ exists iff $\sigma(\psi) \subseteq \sigma(\phi)$, i.e. $V_\psi \subseteq V_\phi$. Two abstractions are isomorphic in $\text{Abs}(\mathcal{X})$ iff they are equivalent in the sense of theorem 2.2. Thus the defect $\delta(\cdot, \mathcal{F}) : \text{Abs}(\mathcal{X}) \rightarrow [0, \infty]$ is a *functor-invariant* (constant on isomorphism classes) and is *monotone*: $\phi \rightarrow \psi$ implies $\delta(\psi, \mathcal{F}) \geq \delta(\phi, \mathcal{F})$, because $V_\psi \subseteq V_\phi$ shrinks the projection target. This monotonicity is the seed of the composability obstruction (theorem 6.1).

The lattice of subspaces $\{V_\phi\}$ ordered by inclusion is the state space of the whole theory: **A1** asks V_ϕ to be small, **A2** asks $\mathcal{F}'$ to lie in V_ϕ , the defect asks $\mathcal{F}$ to lie in V_ϕ , and **A3** is a chain in this lattice. We now work through each axiom in these terms.

3. COMPRESSION: THE DEFECT FRONTIER IS A NONLINEAR WIDTH

Axiom A1 trades defect against the size of $\mathcal{Z}$. The clean question is the frontier

$$(4) \quad \delta_m^*(\mathcal{F}) := \inf_{\phi: \mathcal{X} \rightarrow \mathbb{R}^m} \inf_{g \in \mathcal{G}} \sup_{f \in \mathcal{F}} \|f - g \circ \phi\|$$

as a function of the representation dimension $m = \dim \mathcal{Z}$ and the decoder budget $\mathcal{G}$. This is, verbatim, a *nonlinear m -width* of $\mathcal{F}$ in the sense of DeVore–Howard–Micchelli [2, 3]. Two regimes must be separated, and separating them is the whole content of A1.

3.1. Unconstrained decoders degenerate.

Theorem 3.1 (Degeneracy without a decoder budget). *Let μ be non-atomic. Then for every separable $\mathcal{F} \subset L^2(\mu)$ there is a scalar feature map $\phi: \mathcal{X} \rightarrow \mathbb{R}$ ($m = 1$) with $\delta(\phi, \mathcal{F}) = 0$. Hence the unconstrained frontier (4) with $\mathcal{G} = L^2$ collapses: $\delta_1^*(\mathcal{F}) = 0$ for all $\mathcal{F}$.*

Proof. A non-atomic standard probability space is Borel-isomorphic to $([0, 1], \text{Leb})$, so there is a measurable bijection (mod null sets) $\phi: \mathcal{X} \rightarrow [0, 1]$ with $\sigma(\phi)$ the full Borel σ -algebra; then $V_\phi = L^2(\mu)$, $P_\phi = I$, and $\delta(\phi, \mathcal{F}) = 0$ by (3). A single real coordinate already generates the whole σ -algebra because $\mathbb{R}$ and $\mathbb{R}^d$ have the same Borel cardinality; the decoder g that inverts the (wildly discontinuous) coding carries all the complexity. $\square$

Theorem 3.1 is not a defect of the framework but its first lesson: *a representation is only as good as the regularity of the decoder it admits*. Dimension of $\mathcal{Z}$ is meaningless without a budget on g . This is precisely why the parent paper’s architecture [1, §2] fixes a smooth ($\phi'' \in L^\infty$) decoder of bounded capacity; the smoothness is load-bearing, not cosmetic.

3.2. With a decoder budget: the genuine width and the recovery of d_{loc}^* . We therefore fix a decoder class $\mathcal{G}$ with a complexity budget—canonically $\mathcal{G} = \mathcal{G}_\Lambda := \{g : \text{Lip}(g) \leq \Lambda\}$ (Lipschitz decoders) or a fixed finite-capacity hypothesis class (e.g. a width-bounded network). Write $\delta_m^*(\mathcal{F}; \mathcal{G})$ for the constrained frontier.

Theorem 3.2 (Compression frontier = constrained nonlinear width). *For any decoder budget $\mathcal{G}$ closed under post-composition with 1-Lipschitz maps,*

$$(5) \quad \delta_m^*(\mathcal{F}; \mathcal{G}) = d_m^{\text{man}}(\mathcal{F}; \mathcal{G}) := \inf_{\phi: \mathcal{X} \rightarrow \mathbb{R}^m} \sup_{g \in \mathcal{G}} \|f - g \circ \phi\|_{L^2(\mu)},$$

the manifold (m -dimensional, $\mathcal{G}$ -decoded) width of $\mathcal{F}$. In particular the abstraction frontier is sandwiched between the linear and nonlinear Kolmogorov widths,

$$(6) \quad d_m^{\text{nonlin}}(\mathcal{F}) \leq \delta_m^*(\mathcal{F}; \mathcal{G}) \leq d_m^{\text{lin}}(\mathcal{F}),$$

and for a Sobolev ball $\mathcal{F} = \{\|f\|_{\mathcal{H}^s(\mathcal{M})} \leq R\}$ on a d^ -manifold the two widths coincide at rate m^{-s/d^*} , so*

$$(7) \quad \delta_m^*(\mathcal{F}; \mathcal{G}) \asymp m^{-s/d^*}, \quad (\text{squared}) \quad \asymp m^{-2s/d^*},$$

recovering the parent paper’s lazy rate $\alpha = 2s/d^$ and the identity $\delta_m^{*2} \asymp m^{-2s/d_{\text{loc}}^*}$ with $d_{\text{loc}}^* = d^*$.*

Proof. The first identity is the definition (4) unwound: by theorem 2.1 the inner $\inf_g$ over the *restricted* class $\mathcal{G}$ is the $\mathcal{G}$ -decoded approximation, and the outer infima over ϕ and g are exactly the manifold width. The sandwich is the classical relation between linear, manifold, and nonlinear widths [3]: a manifold method with continuous decoder cannot beat the best m -term nonlinear approximation, nor be worse than the best m -dimensional linear subspace decoded by the identity. The Sobolev rate and the linear/nonlinear coincidence on Sobolev balls are [1, Thm. “width”] read through this identity: the Laplace–Beltrami eigenbasis is rate-optimal, so sorting buys no exponent and $d_{\text{loc}}^* = d^*$. $\square$

Corollary 3.3 (A1 is a rate, not a dimension). *Define the effective dimension of $\mathcal{F}$ as the decay rate of its compression frontier: $d_{\text{loc}}^*(\mathcal{F}) := 2s/\theta$ where $\delta_m^*(\mathcal{F}; \mathcal{G})^2 \asymp m^{-\theta}$. Then A1 (“ $\dim \mathcal{Z} \ll d$ ”) is the statement $d_{\text{loc}}^*(\mathcal{F}) \ll d$, i.e. a fast frontier, not a literal small ambient dimension. For Sobolev $d_{\text{loc}}^* = d^*$; for compositional $\mathcal{F}$, section 6 gives $d_{\text{loc}}^* = \max_i d_i < d^*$; for Besov $p < 2$ the rate is unchanged but the lazy decoder is suboptimal ([1, Rem. “Besov adaptivity gain”]). The frontier rate is the only invariant A1 controls.*

This is the precise sense in which the parent paper’s d_{loc}^* is the A1 content of an abstraction: it is the exponent of the defect–compression frontier, an invariant of the *class*, computed by the best abstraction.

4. REUSABILITY, MADE PRECISE

Axiom A2 as literally stated—learning from $\mathcal{Z}$ no slower than from $\mathcal{X}$ for *all* $f' \in \mathcal{F}' \supseteq \mathcal{F}$ —is false unless ϕ is invertible.

Proposition 4.1 (A2 cannot hold for an enlarged class). *If $V_\phi \neq L^2(\mu)$ then there is $f' \in L^2(\mu)$ with $\inf_g \|f' - g \circ \phi\| > 0$, so no estimator using only ϕ can drive its risk on f' below the bias floor $\|(I - P_\phi)f'\|$, whereas an estimator on $\mathcal{X}$ has no such floor. Hence A2 fails on any $\mathcal{F}'$ containing such an f' .*

Proof. Take any $f' \notin V_\phi$ (exists since V_ϕ is proper closed); by theorem 2.1 the best decoder leaves residual $\|(I - P_\phi)f'\| > 0$, which lower-bounds the L^2 risk of *any* function of ϕ , while a consistent estimator on $\mathcal{X}$ achieves risk $\rightarrow 0$. $\square$

The repair is to read A2 not as a universal but as a *characterisation of the family it holds for*. The correct object is the cone of tasks the abstraction represents.

Definition 4.2 (Transfer-compatible cone). The ϵ -transfer cone of ϕ is $\text{Cone}_\epsilon(\phi) := \{f' \in L^2(\mu) : \|(I - P_\phi)f'\| \leq \epsilon\}$, the tasks ϕ represents to tolerance ϵ . For $\epsilon = 0$ this is the subspace V_ϕ itself.

Theorem 4.3 (Reusability holds exactly on the cone). *Let learning be regularised least squares with the parent paper’s spectral rates, and let $f' \in V_\phi$. Then learning f' from the abstraction $\mathcal{Z}$ is no slower than from $\mathcal{X}$, and is strictly faster whenever $d_{\text{loc}}^*(V_\phi) < d^*$: the estimation rate from $\mathcal{Z}$ is governed by the source exponent of f' relative to the learned kernel on V_ϕ , whose capacity exponent is $b_{V_\phi} = 2s/d_{\text{loc}}^*(V_\phi) \geq b_0$, yielding minimax rate $D^{-\beta_{V_\phi}}$ with $\beta_{V_\phi} = \frac{2s}{2s+d_{\text{loc}}^*(V_\phi)} \geq \beta_0(d^*)$. For $f' \in \text{Cone}_\epsilon(\phi) \setminus V_\phi$ the rate is the same up to an additive, irreducible bias ϵ^2 :*

$$(8) \quad \inf_{\hat{f} \in V_\phi} \mathbb{E} \left\| \hat{f} - f' \right\|^2 \leq \underbrace{C D^{-\beta_{V_\phi}}}_{\text{estimation, on } V_\phi} + \underbrace{\|(I - P_\phi)f'\|^2}_{\leq \epsilon^2, \text{ irreducible}}.$$

Proof. On V_ϕ the hypothesis space is a subspace of $L^2(\mu)$ of effective capacity exponent $b_{V_\phi} = 2s/d_{\text{loc}}^*(V_\phi)$ (theorem 3.3): the nuisance directions orthogonal to V_ϕ are removed, so the effective

dimension of the estimation problem is $d_{\text{loc}}^*(V_\phi) \leq d^*$, never larger. The standard regularised-least-squares minimax rate $D^{-2s/(2s+d_{\text{eff}})}$ [6] is monotone decreasing in d_{eff} , giving $\beta_{V_\phi} \geq \beta_0(d^*)$: no slower, strictly faster when $d_{\text{loc}}^*(V_\phi) < d^*$. For general f' , decompose $f' = P_\phi f' + (I - P_\phi)f'$; the estimator on V_ϕ targets $P_\phi f'$ (rate as above), and the orthogonal complement contributes a fixed bias $\|(I - P_\phi)f'\|^2 \leq \epsilon^2$ by definition of the cone, additively by the Pythagorean identity. $\square$

Remark 4.4 (A2 restated). Theorem 4.3 gives the precise form of A2: *reusability is not a property of ϕ alone but of the pair $(\phi, \mathcal{F}')$, and it holds iff $\mathcal{F}' \subseteq \text{Cone}_\epsilon(\phi)$, degrading by exactly the projection residual outside the cone.* The “enlarged family” of the original A2 is forced to be the transfer cone—which is the bridge to P4: transfer is the question of when a *second* class lands in the first’s cone.

5. TRANSFER BETWEEN CONCEPT CLASSES (P4)

Theorem 4.4 reduces transfer to a geometric question: given the abstraction $\phi = \phi_{\mathcal{F}_1}$ built for $\mathcal{F}_1$, how far does $\mathcal{F}_2$ stick out of its subspace $V_1 = V_\phi$? We make this a metric on classes and prove the bound P4 requested.

For a bounded class $\mathcal{F}$ let $V_n(\mathcal{F})$ denote a width-optimal n -dimensional subspace (achieving $d_n^{\text{lin}}(\mathcal{F})$), $P_n(\mathcal{F})$ its projection, and $R(\mathcal{F}) := \sup_{f \in \mathcal{F}} \|f\|$. For two equidimensional subspaces U, W ($\dim U = \dim W = n$) the *gap* is the standard quantity [7, Kato]

$$(9) \quad \Theta(U, W) := \|P_U - P_W\| = \|(I - P_U)P_W\| = \sin \theta_{\max}(U, W) \in [0, 1],$$

the sine of the largest principal angle; Θ is a metric on the Grassmannian $\text{Gr}(n, L^2)$.

Definition 5.1 (Distance between concept classes). At resolution n , define

$$(10) \quad \text{dist}_n(\mathcal{F}_1, \mathcal{F}_2) := \Theta(V_n(\mathcal{F}_1), V_n(\mathcal{F}_2)) + \frac{d_n^{\text{lin}}(\mathcal{F}_1) + d_n^{\text{lin}}(\mathcal{F}_2)}{R(\mathcal{F}_1) \vee R(\mathcal{F}_2)}.$$

The first term is the misalignment of the width-optimal subspaces; the second penalises classes not yet resolved at level n . dist_n is symmetric, vanishes when the classes share a width-optimal subspace and are n -resolved, and satisfies the triangle inequality up to the d_n terms (it is a genuine metric on n -resolved classes).

Theorem 5.2 (Transfer bound). *Let $\phi = \phi_{\mathcal{F}_1}$ be an abstraction whose subspace is width-optimal for $\mathcal{F}_1$ at resolution n , $V_1 = V_n(\mathcal{F}_1)$. Then*

$$(11) \quad \boxed{\delta(\phi_{\mathcal{F}_1}, \mathcal{F}_2) \leq d_n^{\text{lin}}(\mathcal{F}_2) + R(\mathcal{F}_2) \Theta(V_n(\mathcal{F}_1), V_n(\mathcal{F}_2)) \leq C(\mathcal{F}_2) \text{dist}_n(\mathcal{F}_1, \mathcal{F}_2),}$$

with $C(\mathcal{F}_2) = R(\mathcal{F}_2)$. In words: the transfer defect is bounded by the intrinsic n -width of the new class plus its radius times the misalignment of the two classes’ optimal subspaces. This is the inequality of P4, with $\text{dist} = \text{dist}_n$.

Proof. Fix $f \in \mathcal{F}_2$ and write $V_2 = V_n(\mathcal{F}_2)$, $P_i = P_{V_i}$. Then

$$\|(I - P_1)f\| \leq \|(I - P_1)P_2f\| + \|(I - P_1)(I - P_2)f\| \leq \|(I - P_1)P_2\| \|f\| + \|(I - P_2)f\|.$$

The first factor is $\Theta(V_2, V_1) = \Theta(V_1, V_2)$ (symmetry of the gap for equidimensional subspaces); $\|f\| \leq R(\mathcal{F}_2)$; and $\|(I - P_2)f\| \leq d_n^{\text{lin}}(\mathcal{F}_2)$ since V_2 is width-optimal for $\mathcal{F}_2$. Taking $\sup_{f \in \mathcal{F}_2}$ and using (3) gives the first inequality; the second is theorem 5.1 with $C = R(\mathcal{F}_2)$. $\square$

Remark 5.3 (This is transfer learning as a theorem). Theorem 5.2 is the statement that empirical transfer-learning practice has lacked as a theorem: a representation learned for $\mathcal{F}_1$ transfers to $\mathcal{F}_2$ to the extent that the two tasks share principal directions (Θ small) and the new task is

already simple in its own right ($d_n(\mathcal{F}_2)$ small). The two failure modes—“the new task needs features the old one never saw” ($\Theta \approx 1$) and “the new task is intrinsically hard” ($d_n(\mathcal{F}_2)$ large)—are separated and quantified. The bound is tight in the sense that taking $\mathcal{F}_2$ a single unit vector $f \perp V_1$ gives $\delta = 1 = \Theta \cdot R$, saturating the second term.

Remark 5.4 (Towards the geometry of task space). dist_n makes the space of n -resolved concept classes a metric space, fibred over the Grassmannian $\text{Gr}(n, L^2)$ by the map $\mathcal{F} \mapsto V_n(\mathcal{F})$. Transfer is small along geodesics of this fibration. This is the first concrete instance of the parent note’s “geometry of task space”; its completion—a Riemannian or Finsler structure on $\text{Gr}(n, L^2)$ under which dist_n is the induced distance and along whose geodesics curricula are optimal—is theorem 13.1.

6. COMPOSABILITY: AN OBSTRUCTION AND ITS RESOLUTION (A3, P2)

A3 stacks abstractions. In the lattice of subspaces (theorem 2.3) a stack is a chain, and the first fact is negative.

Theorem 6.1 (Naive composition can only raise the defect). *Let $\phi_1 : \mathcal{X} \rightarrow \mathcal{Z}_1$ and $\phi_2 : \mathcal{Z}_1 \rightarrow \mathcal{Z}_2$, and write $\Phi = \phi_2 \circ \phi_1$. Then for every concept class $\mathcal{F} \subset L^2(\mu)$,*

$$(12) \quad V_\Phi \subseteq V_{\phi_1} \quad \implies \quad \delta(\Phi, \mathcal{F}) \geq \delta(\phi_1, \mathcal{F}).$$

More generally, along any chain $\mathcal{X} \xrightarrow{\phi_1} \mathcal{Z}_1 \xrightarrow{\phi_2} \dots \xrightarrow{\phi_L} \mathcal{Z}_L$ the defects against a fixed $\mathcal{F}$ are non-decreasing in depth: $\delta(\phi_1, \mathcal{F}) \leq \delta(\phi_2 \circ \phi_1, \mathcal{F}) \leq \dots \leq \delta(\Phi_L, \mathcal{F})$.

Proof. $\Phi = \phi_2 \circ \phi_1$ is $\sigma(\phi_1)$ -measurable, so $\sigma(\Phi) \subseteq \sigma(\phi_1)$ and $V_\Phi \subseteq V_{\phi_1}$. Projection residual onto a smaller subspace is larger: $\|(I - P_\Phi)f\| \geq \|(I - P_{\phi_1})f\|$ for all f (Pythagoras). Take $\sup_{f \in \mathcal{F}}$. Iterating along the chain gives monotonicity. $\square$

Theorem 6.1 says that the literal reading of A3—“defect smaller at each deeper layer, all measured against the same $\mathcal{F}$ ”—is *impossible*: stacking coarsens, and coarsening against a fixed target can only lose. Empirically this is the statement that a deep stack of bottlenecks, each trying to reconstruct the *same* output, degrades; what rescues depth is that each layer is responsible for a *different* task. The resolution is therefore not a single $\mathcal{F}$ but a filtration.

Definition 6.2 (Level filtration and quotient tasks). A *level filtration* of $\mathcal{F}$ adapted to a chain (ϕ_i) is a sequence of concept classes $\mathcal{F} = \mathcal{F}^{(0)}, \mathcal{F}^{(1)}, \dots, \mathcal{F}^{(L)}$ together with the requirement that $\mathcal{F}^{(i)}$ is represented on $\mathcal{Z}_i$ (i.e. $\mathcal{F}^{(i)} \subseteq V_{\phi_i}$ pushed to $\mathcal{Z}_i$) and that the *residual task* at level i ,

$$\mathcal{R}^{(i)} := \{ (I - P_{\phi_{i-1}})f : f \in \mathcal{F}^{(i-1)} \},$$

is what ϕ_i must newly represent. The chain is *well-composed to tolerance ϵ* if $\sup_i \delta(\phi_i, \mathcal{R}^{(i)}) \leq \epsilon$ level by level.

Under this reading the parent paper’s compositional theorem supplies the existence half of A3.

Proposition 6.3 (Parent compositional theorem = existence on $\mathcal{C}_{d_{\text{loc}}}^s$). *Let $\mathcal{F} = \mathcal{C}_{d_{\text{loc}}}^s$ be the compositional Sobolev class of targets $f^* = g_L \circ \dots \circ g_1$ with each constituent $g_i \in \mathcal{H}^s$ depending on an intermediate representation of intrinsic dimension d_i . Then the chain of constituents is a level filtration (theorem 6.2) in which ϕ_i realises g_i , each level defect obeys $\delta(\phi_i, \mathcal{R}^{(i)}) \asymp m_i^{-s/d_i}$ at width m_i , and the composite attains the rate m^{-2s/d_{loc}^*} with $d_{\text{loc}}^* = \max_i d_i$. The existence of the well-composed chain holds whenever each constituent is $\mathcal{H}^s$ and the per-level interaction gap is positive, $\Delta_i > 0$ ([1, Thm. “comp”, Thm. “hleap”]).*

Proof (reduction). By theorem 3.2 each level’s defect is the manifold width of an $\mathcal{H}^s$ class in its own dimension d_i , hence $\asymp m_i^{-s/d_i}$. The error-propagation bound of [1, Thm. “comp”] (bounded amplification through smooth constituents) sums the $L = O(1)$ levels, and equalising accuracy makes the hardest level $\max_i d_i$ bind, giving $d_{\text{loc}}^* = \max_i d_i$ exactly as [1, Thm. “comp”]. The per-level metastability analysis [1, Thm. “hlep”] supplies the $\Delta_i > 0$ condition under which the filtration is realised by training rather than merely existing. $\square$

This is genuine progress on P2 for one class. The general existence question—which $\mathcal{F}$ admit a well-composed chain at all—remains open, and we state it precisely.

Conjecture 6.4 (Existence of composable abstractions, P2 — partially promoted). A class $\mathcal{F} \subset L^2(\mu)$ admits a depth- L well-composed chain of per-level defect $\leq \epsilon$ and per-level dimension $d_1 \geq \dots \geq d_L$ iff its width frontier is *multiscale-factorisable*: there exist closed subspaces $L^2(\mu) \supseteq W_1 \supseteq \dots \supseteq W_L$ and residual classes $\mathcal{R}^{(i)} \subset W_{i-1} \ominus W_i$ such that (a) $\sup_i d_{\dim}(\mathcal{R}^{(i)}) \leq \epsilon$ at the prescribed dimensions, and (b) the recomposition map $\bigoplus_i \mathcal{R}^{(i)} \rightarrow \mathcal{F}$ is bounded below (no destructive interference across levels). The interaction gaps Δ_i of the parent paper are conjectured to equal the inf-gaps $\inf \sigma(W_{i-1} \ominus W_i\text{-restricted Gram})$ of clause (b); thus $\Delta_i > 0$ matches clause (b), and the parent’s compositional class is the case where (a) and (b) hold by construction.

Status: The *necessity* direction is a theorem for $\mathcal{F}_{s,p}^{\text{Bsv}}$ with $p < 2$ (theorems 6.10 and 6.11): fast nonlinear width does not imply serial factorisability. On the *index-chain class*, both directions hold and reduce to the single inequality $\Delta_{\min} > 0$ (theorems 11.4 and 11.5). What remains at conjecture grade is the *general* necessity (whether every class resisting serialisation does so via a $c_{\text{recomp}} \rightarrow 0$ collapse) and the recomposition bound for nonlinear constituents.

Remark 6.5 (Scope of the conjecture). Theorem 6.4 is falsifiable in its hard direction: a class with a fast width frontier but no factorisation of the frontier into level-residuals would have a small *shallow* defect yet no efficient *deep* representation—separating “compressible” from “hierarchically compressible.” The Besov $p < 2$ case of [1, Rem. “Besov adaptivity gain”] is suggestive: it is compressible (adaptive width beats linear) but *not* via composition (the parameter-rate stays $2s/d^*$), so it should *fail* clause (b) while passing a naive reading of (a)—a concrete test of the conjecture’s necessity direction.

6.1. The necessity direction: a Besov counterexample. We now prove the necessity direction of theorem 6.4: a class can have an *arbitrarily fast nonlinear width frontier* (clause (a) easily satisfied at every level) yet be *incapable* of supporting a well-composed serial chain of positive depth, because every candidate filtration forces the recomposition constant $c_{\text{recomp}} \rightarrow 0$. The construction is an explicit *adaptive-wavelet* class in a Besov space with $p < 2$.

6.1.1. The adaptive-wavelet class in $\mathcal{B}_{p,q}^s$. Fix dimension $d = 1$ for clarity (the construction extends to $d \geq 2$ by tensorisation). Let $\psi \in C^r(\mathbb{R})$ be a Daubechies mother wavelet of order $r > s$ (compactly supported, r vanishing moments). The *wavelet system* at resolution $j \in \mathbb{N}$ and location $k \in \mathbb{Z}$ is $\psi_{j,k}(x) := 2^{j/2}\psi(2^j x - k)$, forming an orthonormal basis of $L^2([0, 1])$. For $0 < p < 2$, $q = \infty$, and $s > 1/p - 1/2$ (so that the Besov embedding $\mathcal{B}_{p,\infty}^s \hookrightarrow L^2$ holds), define

Definition 6.6 (Sparse-wavelet class).

$$(13) \quad \mathcal{F}_{s,p}^{\text{Bsv}} := \left\{ f = \sum_{j=0}^{\infty} \sum_{k=0}^{2^j-1} c_{j,k} \psi_{j,k} : \left(\sum_{j \geq 0} 2^{j(s-1/2+1/p)p} \left(\sum_k |c_{j,k}|^p \right) \right)^{1/p} \leq 1 \right\}.$$

This is the unit ball of $\mathcal{B}_{p,\infty}^s$ in the wavelet characterisation. For any $f \in \mathcal{F}_{s,p}^{\text{Bsv}}$ the *support pattern* $\Lambda(f) := \{(j, k) : c_{j,k} \neq 0\}$ is at most ℓ^p -sparse at each scale j .

Remark 6.7 (The key asymmetry: $p < 2$ vs $p = 2$; corrected width formula). For $p = 2$ the class is the Sobolev ball $\mathcal{H}^s$ and the optimal m -term approximation is the fixed linear truncation to top Laplace–Beltrami eigenfunctions; the nonlinear and linear widths coincide at m^{-s} . For $p < 2$ the coefficients are ℓ^p -sparse—at each resolution j only $O(2^{j(1-p/2)})$ of the 2^j coefficients are large—so the optimal approximation is *adaptive thresholding* (select the m globally largest coefficients). Setting $\sigma := 1/p - 1/2 > 0$, the two widths satisfy

$$(14) \quad d_m^{\text{nonlin}}(\mathcal{F}_{s,p}^{\text{Bsv}}) \asymp m^{-s}, \quad d_m^{\text{lin}}(\mathcal{F}_{s,p}^{\text{Bsv}}) \asymp m^{-(s-\sigma)},$$

so nonlinear beats linear by a gap exponent $\sigma > 0$ (larger σ = sparser = larger gap). The formula $m^{-(s-\sigma)}$ for the linear width follows from (a) the Besov embedding $\mathcal{B}_{p,\infty}^s \hookrightarrow H^{s-\sigma}$ and (b) the Sobolev linear width $m^{-(s-\sigma)}$; see [3]. (*Note: an earlier version of this remark wrote $m^{-s(1/p-1/2)^{-1}}$ for the linear width, which is incorrect for all $p \neq 1$. The correct formula is (14).*) The adaptive thresholding is *not linear* and *not hierarchical*: the support $\Lambda(f)$ changes with f .

6.1.2. *Fast nonlinear width: clause (a) is trivially satisfied.*

Lemma 6.8 (Clause (a) holds at every decomposition). *For any partition of scales $J = J_1 \sqcup \dots \sqcup J_L$ (each J_i a set of resolution levels), define the level- i projection*

$$P_i f := \sum_{j \in J_i} \sum_k c_{j,k}(f) \psi_{j,k},$$

the level- i residual class $\mathcal{R}^{(i)} := \{P_i f : f \in \mathcal{F}_{s,p}^{\text{Bsv}}\}$, and the per-level width

$$\varepsilon_i(m_i) := d_{m_i}^{\text{nonlin}}(\mathcal{R}^{(i)}).$$

Then for any $\varepsilon > 0$ one can choose the partition and budgets $\{m_i\}$ with $\sum_i m_i = m$ such that $\sup_i \varepsilon_i(m_i) \leq \varepsilon$ at total budget $m = O(\varepsilon^{-1/s})$, i.e. matching the global nonlinear width rate.

Proof. The m -term nonlinear approximation of $\mathcal{F}_{s,p}^{\text{Bsv}}$ is achieved by the *thresholding estimator*: select the m largest-magnitude wavelet coefficients. Since the Besov condition (13) bounds the ℓ^p -norm of coefficients at each level, the error of truncating to m terms is $\asymp m^{-s}$ [3]. Partitioning scales into $L = O(\log(1/\varepsilon)/\log 2)$ dyadic bands and allocating the budget proportionally to the energy at each band, the per-level error is $\leq \varepsilon$ and the total budget is $m = \sum_i m_i = O(\varepsilon^{-1/s})$, matching the global rate. Hence clause (a) of theorem 6.4 is satisfied with equality at the optimal nonlinear rate. $\square$

6.1.3. *Precise definition of the recomposition constant.*

Definition 6.9 (Recomposition constant). Let $(\phi_i, E_i)_{i=1}^L$ be a *linear* serial filtration: a chain of closed orthogonal subspaces $E_i := W_{i-1} \oplus W_i$ with $L^2 \supseteq W_1 \supseteq \dots \supseteq W_L$ and $\mathcal{A} := \bigoplus_{i=1}^L E_i$. Define

$$(15) \quad \varepsilon_{\text{chain}}(\mathcal{F}, m) := \sup_{f \in \mathcal{F}} \|f - P_{\mathcal{A}} f\|_{L^2(\mu)}, \quad \varepsilon_{\text{opt}}(\mathcal{F}, m) := d_m^{\text{nonlin}}(\mathcal{F}).$$

The *recomposition constant* is

$$(16) \quad c_{\text{recomp}}(\mathcal{F}, m, \{E_i\}) := \frac{\varepsilon_{\text{opt}}(\mathcal{F}, m)}{\varepsilon_{\text{chain}}(\mathcal{F}, m)} \in (0, 1].$$

$c_{\text{recomp}} = 1$: the chain achieves the nonlinear-optimal error (well-composed). $c_{\text{recomp}} \rightarrow 0$: the chain degrades relative to the nonlinear optimum at the same budget.

6.1.4. *The recomposition constant collapses: clause (b) fails.*

Theorem 6.10 (Necessity: sparse-wavelet classes are not serially factorisable, corrected). *Let $\sigma := 1/p - 1/2$, $s > \sigma > 0$, and $L = O(1)$. For every linear serial filtration (fixed orthogonal subspace at each level) of total budget m and concept class $\mathcal{F}_{s,p}^{\text{Bsv}}$:*

- (i) Budget excess. *To achieve per-level defect $\leq \varepsilon$ via fixed subspaces, each level budget must satisfy*

$$(17) \quad m_i \geq c(s, p) \varepsilon^{-1/(s-\sigma)},$$

exceeding the nonlinear budget $\varepsilon^{-1/s}$ by a factor $\varepsilon^{-\sigma/(s(s-\sigma))} \rightarrow \infty$.

- (ii) Recomposition collapse. *At nonlinear budget $m_{\text{opt}} := \varepsilon^{-1/s}$:*

$$(18) \quad c_{\text{recomp}}(\mathcal{F}_{s,p}^{\text{Bsv}}, m_{\text{opt}}, \{E_i\}) \leq C(s, p) \varepsilon^{\sigma/s} \rightarrow 0 \quad (\sigma > 0).$$

- (iii) Non-factorisability. *No fixed-subspace serial chain achieves simultaneously: per-level defect at the nonlinear-optimal budget and $c_{\text{recomp}} \geq c_0 > 0$ uniformly.*

Proof. (i) Budget excess. A fixed m_i -dimensional subspace E_i achieves best-case per-level defect equal to the linear m_i -width of $\mathcal{R}^{(i)} \subset \mathcal{F}_{s,p}^{\text{Bsv}}$. By the Besov embedding $\mathcal{B}_{p,\infty}^s \hookrightarrow H^{s-\sigma}$ and the Sobolev linear-width formula:

$$(19) \quad d_{m_i}^{\text{lin}}(\mathcal{R}^{(i)}) \geq c m_i^{-(s-\sigma)}.$$

For this to be $\leq \varepsilon$: $m_i \geq \varepsilon^{-1/(s-\sigma)} \gg \varepsilon^{-1/s}$ since $1/(s-\sigma) > 1/s$ for $\sigma > 0$. $\square_{(i)}$

(ii) Recomposition collapse. At nonlinear budget $m_{\text{opt}} = \varepsilon^{-1/s}$, the reconstruction space $\mathcal{A}$ is a linear m_{opt} -dimensional subspace of L^2 . By (14):

$$(20) \quad \varepsilon_{\text{chain}} \geq d_{m_{\text{opt}}}^{\text{lin}} \asymp m_{\text{opt}}^{-(s-\sigma)} = \varepsilon^{(s-\sigma)/s}, \quad \varepsilon_{\text{opt}} \asymp m_{\text{opt}}^{-s} = \varepsilon.$$

Therefore $c_{\text{recomp}} = \varepsilon_{\text{opt}}/\varepsilon_{\text{chain}} \leq \varepsilon/\varepsilon^{(s-\sigma)/s} = \varepsilon^{\sigma/s} \rightarrow 0$. $\square_{(ii)}$

(iii) From (i) and (ii): at the nonlinear budget $\varepsilon_{\text{chain}} \gg \varepsilon$ (clause a fails); to satisfy clause (a) requires super-optimal budget, at which the linear-rate constraint forces $c_{\text{recomp}} \rightarrow 0$ unless $\sigma = 0$ ($p = 2$). $\square_{(iii)}$ $\square$

Corollary 6.11 (Necessity direction is a theorem for fixed-subspace chains). *For $\mathcal{F}_{s,p}^{\text{Bsv}}$ with $p < 2$: fast nonlinear width does not imply serial factorisability for fixed-subspace chains. The gap is the sparsity index $\sigma = 1/p - 1/2 > 0$.*

Remark 6.12 (Scope: linear vs. adaptive encoders). Theorem 6.10 is for *fixed-subspace* (linear) encoders. If each level encoder is adaptive (data-dependent thresholding), an adaptive serial chain can mimic global thresholding level by level and achieve the nonlinear rate with $c_{\text{recomp}} = 1$. The non-trivial non-factorisability content is exclusively the linear/fixed-subspace setting.

Conjecture 6.13 (Sharper rate for adaptive chains). For adaptive (nonlinear) serial encoders satisfying the consistency conditions of theorem 6.4, the collapse rate is $c_{\text{recomp}} \leq C\varepsilon^{2\sigma}$ ($\sigma = 1/p - 1/2$), achieved when pairs can be placed at the same resolution with orthogonal supports matching the level budget exactly. This tighter rate requires an information-theoretic argument beyond the linear-width gap and is left as a conjecture.

Remark 6.14 (What the counterexample says about deep learning). $\mathcal{F}_{s,p}^{\text{Bsv}}$ is “compressible but not hierarchically compressible” via fixed subspaces. Optimal representation requires adaptive coefficient selection, which varies with the input—a fundamentally parallel operation. A fixed-subspace serial chain commits to a subspace before seeing the signal, wasting budget on quiet scales. Architecturally: this explains why attention mechanisms (adaptive, input-dependent thresholding) outperform fixed convolutional hierarchies on variable-sparsity functions—the Besov obstruction is precisely what attention bypasses.

7. LEARNABILITY AND THE STATISTICAL-QUERY BARRIER (P3)

P3 is the hardest: from i.i.d. samples of a *single* task $f \in \mathcal{F}$, build a ϕ satisfying A1–A3. The parent paper gives the two smallest cases as theorems and the obstruction as a barrier; we organise these into a single index whose general theory is the open problem.

Definition 7.1 (Learnability index). For a concept class $\mathcal{F}$ and a feature-construction algorithm $\mathcal{A}$ using D samples, let the *abstraction risk* be $R_D(\mathcal{A}, \mathcal{F}) := \sup_{f \in \mathcal{F}} \mathbb{E} \delta(\hat{\phi}_D, \{f\})^2$ where $\hat{\phi}_D = \mathcal{A}(\text{samples of } f)$. The *learnability index* of $\mathcal{F}$ is the critical sample exponent

$$(21) \quad \kappa(\mathcal{F}) := \sup \left\{ \kappa : \inf_{\mathcal{A}} R_D(\mathcal{A}, \mathcal{F}) \lesssim D^{-\kappa} \right\},$$

the best achievable rate at which an abstraction matching f ’s structure can be *built* (not merely the rate at which f is regressed). $\kappa(\mathcal{F}) > 0$ means the abstraction is learnable from data; $\kappa(\mathcal{F}) = 0$ means it is not learnable at the structural level by any efficient method in the relevant query model.

Proposition 7.2 (The parent paper computes κ at $L = 1, 2$). *For the single-index/shallow class ($L = 1$), $\kappa > 0$ holds whenever the Gram lower bound $\lambda_{\min}(G) \geq \underline{c}$ is met [1, Thm. 8.5]; for the two-level class ($L = 2$), $\kappa > 0$ holds whenever the interaction gap exceeds threshold, $\Delta \geq \Delta_0$ [1, Thm. 9.5]. Conversely, at $\Delta = 0$ the construction is statistical-query hard: no polynomial-query SQ algorithm builds the level-2 abstraction [1, §(barrier)], so $\kappa = 0$ there.*

The picture is a phase transition in the order parameter Δ : above threshold the abstraction is learnable ($\kappa > 0$), at the gap closure it is not ($\kappa = 0$, SQ-hard). The general theory P3 wants is the function $D(\epsilon, \kappa)$ and the identification of $\kappa(\mathcal{F})$ from structural invariants of $\mathcal{F}$ for arbitrary L .

Conjecture 7.3 (Learnability index = minimal interaction gap, P3 — partially promoted). For a class admitting a depth- L well-composed chain (theorem 6.4) with per-level interaction gaps $\Delta_1, \dots, \Delta_L$, the learnability index is controlled by the *smallest* gap:

$$(22) \quad \kappa(\mathcal{F}) > 0 \iff \Delta_{\min} := \min_i \Delta_i > 0,$$

and in the learnable phase the sample complexity to build the chain to per-level tolerance ϵ is $D(\epsilon, \kappa) \asymp \epsilon^{-1/\kappa}$ with $\kappa = \kappa(\Delta_{\min}, \{d_i\})$ a rate that degrades as $\Delta_{\min} \rightarrow 0$ and as $\max_i d_i$ grows. The boundary $\Delta_{\min} = 0$ is SQ-hard by the level-wise reduction to [1, §(barrier)]: a single closed gap hides one level’s structure from all low-degree queries, and the chain cannot be completed past it.

Status: On the *index-chain class*, the equivalence $\kappa(\mathcal{F}) > 0 \Leftrightarrow \Delta_{\min} > 0$ and the SQ boundary are theorems (theorems 11.4 and 11.5). What remains at conjecture grade is the *general* learnability index $\kappa(\mathcal{F})$ for arbitrary concept classes, in particular for classes whose level measures are non-smooth, non-Gaussian, and lack score regularity—a regime that requires tools from stochastic analysis beyond the subspace-geometry framework of this paper.

Remark 7.4 (The precise gap to the general index, and the concrete next step). The proven half of theorem 7.3 uses score regularity in exactly one place: to turn the level read-out into the second-order score tensor (theorems 9.3 and 11.4), which requires the level measure to admit a second-order Stein operator. Every measure with a *Stein kernel* τ_ν (equivalently, a Poincaré-regular ν with $\sqrt{\tau_\nu}$ in the Dirichlet domain) falls in this proven class; broadening “score-regular” to “admits a Stein kernel” is therefore not new content—it is the same subspace-geometry argument with a weaker regularity label, and we do *not* record it as progress. The conjecture’s open content forms the complement: measures with *no* Stein kernel—singular laws, fractal

supports, and heavy-tailed measures with $\mathbb{E} \|x\|^2 = \infty$ —for which the second-order tensor does not exist and the read-out is genuinely nonlinear in f .

Concrete next step (not “tools beyond”): define $\kappa(\mathcal{F})$ for such measures through the *correlational statistical-query* dimension $\text{SQ-dim}_{\text{corr}}(\mathcal{F})$ rather than through a score tensor, and prove the two implications separately: (a) an *upper* bound $\kappa(\mathcal{F}) \geq c/\text{SQ-dim}_{\text{corr}}(\mathcal{F})$ by exhibiting a correlational query algorithm (this does not need a Stein kernel); (b) a matching SQ *lower* bound at $\Delta_{\min} = 0$ by an orthogonalised-function family on the singular support (a Fano/packing construction in $L^2(\nu)$), which is the genuinely open analytic step. Closing (b) for one explicit non-Stein class (e.g. a self-similar measure on a Cantor-type support) would be the first evidence beyond the subspace-geometry frame.

Remark 7.5 (The one structural input that is still assumed). Theorem 7.3 inherits the parent programme’s standing debt: $\Delta_i > 0$ is used as an *input*, derived from the architecture only in the $L \leq 2$ theorems. The decisive next step *inside* P3—identified already in the parent note—is to *derive* $\Delta_i > 0$ from the structure of $\mathcal{F}$ as a theorem rather than assume it. Theorem 6.4 proposes the bridge: Δ_i is the inf-gap of the level- i residual Gram, so a lower bound on Δ_i is a non-degeneracy statement about the multiscale factorisation. Proving that non-degeneracy for a named class beyond $L = 2$ is the concrete first deliverable of P3.

8. DERIVING THE GAP: THE MULTI-INDEX CLASS

We now discharge the standing debt of theorem 7.5: for a named concept class we *compute* the interaction gap Δ from a checkable structural property, turning the input hypothesis of theorem 7.3 into a theorem. We work in the canonical home of the statistical-query barrier—the Gaussian multi-index model—which is a genuinely different class from the parent’s compositional Sobolev one, so the result is new content rather than a re-reading.

8.1. The class and its abstraction.

Definition 8.1 (Multi-index class). Let $\mu = N(0, I_d)$ and

$$(23) \quad \mathcal{F}_{k,R}^{\text{idx}} := \left\{ f^*(x) = g(U^\top x) : U \in \mathbb{R}^{d \times k}, U^\top U = I_k, \|g\|_{\mathcal{H}^s(\mathbb{R}^k)} \leq R \right\}.$$

The abstraction to be *learned* is the linear feature map $\phi(x) = U^\top x$, i.e. the k -dimensional subspace $\text{col}(U) \subseteq \mathbb{R}^d$; the decoder is the link g . This is the two-level chain $\mathbb{R}^d \xrightarrow{\phi} \mathbb{R}^k \xrightarrow{g} \mathbb{R}$ in the language of A3, with level-1 dimension $d_1 = k \ll d$ (so A1 holds with $d_{\text{loc}}^* = k$) and the link as level 2.

The defect is zero by construction for the right U ; the content is whether U can be *built* from samples (P3), which the next results tie to a derived Δ .

8.2. The gap is the smallest singular value of the link’s expected Hessian. The tool is the Hermite (Stein) tensor. For $x \sim N(0, I_d)$ let $\mathcal{H}_\ell(x) \in (\mathbb{R}^d)^{\otimes \ell}$ be the degree- ℓ Hermite tensor, characterised by the Stein identity $\mathbb{E}[\psi(x) \mathcal{H}_\ell(x)] = \mathbb{E}[\nabla^{\otimes \ell} \psi(x)]$ for smooth ψ of moderate growth [8, 9]. Define the *level- ℓ feature tensor*

$$(24) \quad T_\ell := \mathbb{E}_{x \sim \mu} [f^*(x) \mathcal{H}_\ell(x)] \in (\mathbb{R}^d)^{\otimes \ell},$$

the natural object an order- ℓ method correlates against to expose $\text{col}(U)$.

Lemma 8.2 (The feature tensor factors through the link’s Hermite tensor). *With $C_\ell := \mathbb{E}_{z \sim N(0, I_k)} [\nabla^{\otimes \ell} g(z)] \in (\mathbb{R}^k)^{\otimes \ell}$,*

$$(25) \quad T_\ell = U^{\otimes \ell} C_\ell, \quad \text{hence} \quad \sigma_j(\text{Mat}_1 T_\ell) = \sigma_j(\text{Mat}_1 C_\ell) \quad \forall j,$$

where Mat_1 is the mode-1 unfolding. In particular $\text{col}(\text{Mat}_1 T_\ell) \subseteq \text{col}(U)$, with equality iff $\text{Mat}_1 C_\ell$ has rank k .

Proof. By the Stein identity $T_\ell = \mathbb{E}[\nabla^{\otimes \ell} f^*(x)]$. Since $f^*(x) = g(U^\top x)$ with U constant and linear, the chain rule gives $\nabla_x^{\otimes \ell} f^* = U^{\otimes \ell} (\nabla^{\otimes \ell} g)(U^\top x)$; taking expectations and using $U^\top x \sim N(0, I_k)$ yields $T_\ell = U^{\otimes \ell} C_\ell$. The mode-1 unfolding factors as $\text{Mat}_1 T_\ell = U (\text{Mat}_1 C_\ell) (U^{\otimes (\ell-1)})^\top$; as U and $U^{\otimes (\ell-1)}$ have orthonormal columns, left/right multiplication by them preserves all singular values. The column-space claim follows since U is injective. $\square$

Theorem 8.3 (The interaction gap, derived). *Define the level- ℓ gap $\Delta_\ell := \sigma_k(\text{Mat}_1 T_\ell)$ and let $\ell^* := \min\{\ell : \Delta_\ell > 0\}$ be the information exponent of the class. Then:*

- (i) $\Delta_\ell = \sigma_k(\text{Mat}_1 C_\ell)$ depends only on the link g , not on the hidden U or the ambient d . For $\ell = 2$ this is exactly the smallest singular value of the link’s expected Hessian,

$$(26) \quad \Delta_2 = \sigma_{\min}(\mathbb{E}_{z \sim N(0, I_k)}[\nabla^2 g(z)]).$$

- (ii) $\Delta_{\ell^*} > 0$ iff all k directions of $\text{col}(U)$ are exposed at order ℓ^* , in which case $\text{col}(U)$ is recovered from T_{ℓ^*} by a rank- k SVD, and the abstraction is learnable: the empirical $\hat{T}_{\ell^*}$ concentrates at rate $D^{-1/2} \text{poly}(d^{\ell^*})$, so by the Davis–Kahan bound the recovery error of $\text{col}(U)$ is $\lesssim \Delta_{\ell^*}^{-1} d^{\ell^*/2} D^{-1/2}$ and $\kappa(\mathcal{F}^{\text{idx}}) > 0$ with sample complexity $D \asymp d^{\ell^*} / (\Delta_{\ell^*}^2 \varepsilon^2)$.
- (iii) This identifies the parent paper’s gap: $\Delta = \Delta_2$ is the level-2 instance, so the threshold $\Delta \geq \Delta_0$ of [1, Thm. 9.5] is the curvature condition (26), and theorem 6.4’s identification of Δ_i with a level-residual inf-gap holds here by computation: the residual Gram is the expected Hessian, whose inf-gap is (26).

Proof. (i) is theorem 8.2 with $\text{Mat}_1 C_2 = C_2 = \mathbb{E}[\nabla^2 g]$ (a $k \times k$ symmetric matrix, so its smallest singular value is its k th). (ii): when $\text{Mat}_1 C_{\ell^*}$ has rank k , theorem 8.2 gives $\text{col}(\text{Mat}_1 T_{\ell^*}) = \text{col}(U)$ with spectral gap Δ_{ℓ^*} to the null space; the empirical tensor $\hat{T}_{\ell^*}$ formed from D samples has entrywise fluctuations $O(D^{-1/2})$ scaled by the ℓ^* -th Hermite norms ($\text{poly}(d^{\ell^*})$ entries), and Davis–Kahan converts the gap and the perturbation into the stated subspace error [11, 9]. (iii) is the comparison of definitions. $\square$

Corollary 8.4 (Ellipticity of the link is a checkable sufficient condition for the gap). *If the link satisfies the uniform ellipticity bound $\mathbb{E}_z[\nabla^2 g(z)] \succeq \rho I_k$ (or $\sigma_{\min} \geq \rho$), then $\Delta_2 \geq \rho > 0$ and the abstraction is learnable at the optimal quadratic budget $D \asymp d^2$. Thus the structural restriction*

$$(27) \quad \mathcal{F}_{k,R,\rho}^{\text{idx}} := \{g(U^\top x) \in \mathcal{F}_{k,R}^{\text{idx}} : \sigma_{\min}(\mathbb{E}[\nabla^2 g]) \geq \rho\}$$

forces $\Delta \geq \rho$ over the whole class. This is the deliverable promised in theorem 7.5: $\Delta > 0$ derived from a verifiable property of $\mathcal{F}$ (link curvature), not assumed.

Remark 8.5 (The gap closure is the SQ barrier). If $\Delta_2 = 0$, some unit $w \in \mathbb{R}^k$ has $\mathbb{E}[\nabla^2 g] w = 0$: the direction Uw carries no second-order Hermite signal, so its information exponent is ≥ 3 . Every statistical query of degree < 3 then has vanishing correlation with Uw , and the SQ lower bounds for multi-index recovery [9, 10] give complexity $d^{\Omega(\ell^*)}$, strictly super-quadratic. The parent paper’s “ $\Delta = 0 \Rightarrow \text{SQ-hard}$ ” follows directly: a closed gap hides a direction below the query degree. Theorems 8.3 and 8.5 thus reproduce the phase transition—learnable above the curvature threshold, SQ-hard at its closure—with the order parameter Δ now an explicit functional of the link. (Numerically verified: a link whose Hessian has a zero eigenvalue in one direction yields $\Delta_2 = 0$ there, and that direction is invisible to the level-2 tensor while the curved directions are recovered exactly—see the construction behind (26).)

8.3. Up the chain: the gap is stable, and the recursion closes modulo one named condition. The two-level result is the base case. For depth L the obstacle is that the level- i input is the pushforward $\nu_{i-1} := (\Phi_{i-1})_{\#}\mu$, which is non-Gaussian, and is moreover only known up to the recovery error of the lower levels. Two facts control this; the first is fully general and proved, the second is the single condition the recursion rests on.

Lemma 8.6 (Gap stability under imperfect lower levels). *Let the level- i gap functional be $\Delta_i[\nu] := \sigma_{\min}(\mathbb{E}_{z \sim \nu}[\nabla^2 g_i(z)])$, and suppose g_i has bounded third derivatives, $\|\nabla^3 g_i\|_{\infty} \leq L_i$. If the lower-level abstraction is recovered to accuracy $W_1(\hat{\nu}_{i-1}, \nu_{i-1}) \leq \varepsilon$, then*

$$(28) \quad |\Delta_i[\hat{\nu}_{i-1}] - \Delta_i[\nu_{i-1}]| \leq L_i \varepsilon, \quad \text{so} \quad \Delta_i[\hat{\nu}_{i-1}] \geq \Delta_i[\nu_{i-1}] - L_i \varepsilon.$$

Hence the level- i gap survives the propagation of error from below whenever $\Delta_i > L_i \varepsilon$.

Proof. $z \mapsto \nabla^2 g_i(z)$ is L_i -Lipschitz, so by Kantorovich–Rubinstein $\|\mathbb{E}_{\hat{\nu}}[\nabla^2 g_i] - \mathbb{E}_{\nu}[\nabla^2 g_i]\| \leq L_i W_1(\hat{\nu}, \nu) \leq L_i \varepsilon$; Weyl’s inequality transfers this to the smallest singular value. $\square$

Theorem 8.7 (Multilevel learnability, conditional on a per-level score condition). *Consider the index chain $f^* = g_L \circ \phi_{L-1} \circ \dots$, each level a multi-index step of local dimension d_i with link $g_i \in \mathcal{H}^s$, $\|\nabla^3 g_i\|_{\infty} \leq L_i$. Suppose the per-level identifiability hypothesis (H-score) holds: each intermediate measure ν_{i-1} admits a Stein/score operator under which positivity of the level- i gap $\Delta_i := \Delta_i[\nu_{i-1}] > 0$ suffices to recover the level- i subspace from order-2 correlations (this is theorem 8.3(ii) for $i = 2$, where $\nu_1 = N(0, I_{d_1})$ is Gaussian and the hypothesis is a theorem). Then the chain is learnable level by level, and*

$$(29) \quad \boxed{\Delta_{\min} := \min_i \Delta_i > C \varepsilon_{\max} \implies \kappa(\mathcal{F}) > 0,}$$

where $\varepsilon_{\max}$ is the worst per-level recovery error and $C = \max_i L_i$; conversely if some $\Delta_i = 0$ the chain stalls at level i (SQ-hard there by theorem 8.5). This is theorem 7.3 promoted to a theorem on the index-chain class, modulo (H-score) for $i \geq 3$.

Proof (induction with stability). Base $i = 2$: ν_1 Gaussian, so theorem 8.3(ii) recovers level 2 with error $\varepsilon_2 \lesssim \Delta_2^{-1} d_2^{1/2} D^{-1/2}$. Inductive step: given levels $< i$ recovered with $W_1(\hat{\nu}_{i-1}, \nu_{i-1}) \leq \varepsilon_{<i}$, theorem 8.6 gives $\Delta_i[\hat{\nu}_{i-1}] \geq \Delta_i - L_i \varepsilon_{<i} > 0$ under the displayed condition, so by (H-score) level i is identifiable from order-2 correlations under $\hat{\nu}_{i-1}$, with recovery error controlled by the surviving gap. Composing the $L = O(1)$ steps and using $\Delta_{\min} > C \varepsilon_{\max}$ keeps every gap open, giving a positive overall rate $\kappa > 0$. The converse is theorem 8.5 applied at the first vanishing level. $\square$

Remark 8.8 (What is now a theorem, and the one piece that is not). Theorems 8.3 and 8.4 are unconditional theorems: the two-level gap is the link’s Hessian floor, the SQ phase transition is recovered, and the parent’s assumed threshold is derived. The stability lemma theorem 8.6 is unconditional. The *only* remaining assumption for $L > 2$ is (H-score): that the non-Gaussian intermediate measures admit a score operator under which a positive Hessian-gap suffices for order-2 identifiability. This holds for Gaussian ν_{i-1} (proved), and is expected for log-concave ν_{i-1} with controlled score ([8]); proving it for the specific pushforwards of trained index chains is the precise residue—the same kind of named, isolated hypothesis the parent programme uses, and strictly more specific than the blanket “ $\Delta_i > 0$ ” it replaces. *This residue is discharged in section 9*: theorem 9.3 proves (H-score) for every measure satisfying the score-regularity condition (S), so theorem 8.7 is unconditional on score-regular chains (theorem 9.6).

9. DISCHARGING (H-SCORE): NON-GAUSSIAN INPUTS VIA THE SECOND-ORDER SCORE

Theorem 8.7 rested on one condition, (H-score): that a non-Gaussian intermediate measure admits a Stein operator under which a positive Hessian-gap recovers the level subspace. We now *prove* it under a checkable regularity condition on the measure, removing the last open hypothesis from the index-chain learnability result and, as a bonus, removing the Gaussianity assumption from the base case of theorem 8.3 as well. The device is the *second-order score*, which is to a general measure what the Hermite-2 tensor is to the Gaussian.

9.1. The second-order Stein identity.

Definition 9.1 (Score condition (S), corrected). A probability measure ν on $\mathbb{R}^m$ with density $p \in C^2(\mathbb{R}^m)$, $p > 0$, satisfies condition (S) if the *second-order score* $\mathcal{S}_2 := \nabla^2 p / p$ lies in $L^2(\nu)$, and the following two boundary conditions hold for every $\psi \in \mathcal{H}^s(\mathbb{R}^m)$ and all $i, j \in \{1, \dots, m\}$:

$$(S1) \quad \lim_{R \rightarrow \infty} \int_{\partial B_R} \psi(z) (\partial_i p)(z) dS(z) = 0;$$

$$(S2) \quad \lim_{R \rightarrow \infty} \int_{\partial B_R} (\partial_j \psi)(z) (\partial_i p)(z) dS(z) = 0.$$

Condition (S1) enables the first integration by parts; (S2) enables the second. Both hold for Gaussians, strongly log-concave measures, finite Gaussian mixtures with common covariance, and smooth product measures with super-polynomial tails.

Lemma 9.2 (General second-order Stein identity, corrected proof). *If ν satisfies (S) then for every $\psi \in \mathcal{H}^s(\mathbb{R}^m)$,*

$$(30) \quad \mathbb{E}_\nu[\psi(z) \mathcal{S}_2(z)] = \mathbb{E}_\nu[\nabla^2 \psi(z)], \quad \mathcal{S}_2 = \frac{\nabla^2 p}{p} = \nabla^2 \log p + (\nabla \log p)(\nabla \log p)^\top.$$

For $\nu = N(0, I_m)$ this is the Hermite identity: $\mathcal{S}_2(z) = zz^\top - I = \mathcal{H}_2(z)$.

Proof. For each entry (i, j) , apply integration by parts twice:

$$\begin{aligned} \int \psi \partial_i \partial_j p dz &\stackrel{\text{IBP}_1}{=} - \int (\partial_i \psi) \partial_j p dz + \underbrace{\lim_{R \rightarrow \infty} \int_{\partial B_R} \psi \partial_j p dS}_{=0 \text{ by (S1)}} \\ &\stackrel{\text{IBP}_2}{=} \int (\partial_i \partial_j \psi) p dz - \underbrace{\lim_{R \rightarrow \infty} \int_{\partial B_R} (\partial_i \psi) \partial_j p dS}_{=0 \text{ by (S2)}} = \mathbb{E}_\nu[\partial_i \partial_j \psi]. \end{aligned}$$

Dividing by p and taking the expectation gives $\mathbb{E}_\nu[\psi \partial_i \partial_j p / p] = \mathbb{E}_\nu[\partial_i \partial_j \psi]$. The decomposition $\mathcal{S}_2 = \nabla^2 \log p + (\nabla \log p)(\nabla \log p)^\top$ is the quotient rule; the Gaussian recovery $\mathcal{S}_2 = zz^\top - I$ follows from $\nabla^2 \log p = -I$, $\nabla \log p = -z$. $\square$

9.2. Identifiability with no Gaussianity.

Theorem 9.3 (Non-Gaussian identifiability; (H-score) discharged). *Let the level input $z \sim \nu$ on $\mathbb{R}^m$ satisfy (S), and let $f = g(U^\top z)$ with $U^\top U = I_k$, $g \in \mathcal{H}^s$. Define the score feature tensor $T_2 := \mathbb{E}_\nu[f(z) \mathcal{S}_2(z)]$. Then*

$$(31) \quad T_2 = U C_2 U^\top, \quad C_2 = \mathbb{E}_{z \sim \nu}[\nabla^2 g(U^\top z)] \in \mathbb{R}^{k \times k},$$

so the top- k eigenspace of the symmetric tensor T_2 equals $\text{col}(U)$ whenever $\Delta_2 := \sigma_{\min}(C_2) > 0$, with spectral gap Δ_2 to the null block. From D samples, $\hat{T}_2$ recovers $\text{col}(U)$ at the Davis–Kahan

rate

$$(32) \quad \left\| \hat{P}_U - P_U \right\| \lesssim \frac{\|f \mathcal{S}_2\|_{L^2(\nu)}}{\Delta_2} D^{-1/2},$$

finite by (S). Consequently (H-score) holds for every score-regular ν : identifiability needs only the Hessian-gap $\Delta_2 > 0$, exactly as in the Gaussian case, and the density enters only through the rate via $\|f \mathcal{S}_2\|_{L^2}$.

Proof. By theorem 9.2 with $\psi = f$, $T_2 = \mathbb{E}_\nu[\nabla^2 f(z)]$. Since $f(z) = g(U^\top z)$ with U constant linear, $\nabla_z^2 f = U(\nabla^2 g)(U^\top z)U^\top$; taking expectations gives $T_2 = U C_2 U^\top$. As U has orthonormal columns, the nonzero eigenvalues of T_2 are those of C_2 and the eigenvectors lie in $\text{col}(U)$; a rank- k C_2 ($\Delta_2 > 0$) makes $\text{col}(U)$ the top- k eigenspace with gap Δ_2 to zero. The empirical tensor has fluctuation $\|f \mathcal{S}_2\|_{L^2} D^{-1/2}$ by the CLT entrywise; Davis–Kahan [11] converts gap and fluctuation into the subspace bound. $\square$

Corollary 9.4 (Concrete score-regular classes). *Condition (S) holds, so theorem 9.3 applies, for: the Gaussian (theorem 8.3, recovered as $\mathcal{S}_2 = \mathcal{H}_2$); strongly log-concave ν with $\nabla^2(-\log p)$ bounded and Lipschitz; finite Gaussian mixtures with common covariance; and smooth product measures with sub-exponential tails (e.g. $p \propto e^{-\sum_j V(z_j)}$ with V smooth, V'' bounded below). (Numerically verified on a bimodal Gaussian-mixture input: the score tensor recovers $C_2 = \mathbb{E}[\nabla^2 g]$ to sampling error, the curved directions are recovered while a Hessian-gap-zero direction is correctly invisible—see the construction behind theorem 9.3.)*

Remark 9.5 (The score is learned from *unlabeled* data). $\mathcal{S}_2$ is a property of the input law ν , not of the target f . It is therefore estimable from *unlabeled* samples of z alone—by score matching [12], which fits $\mathcal{S}_2$ without knowing the normalising constant of p —and consumes no label budget. The abstraction-enabling object (the score) is extracted from the data distribution, and labels are used only to read off the link’s curvature through it: self-supervised pre-training followed by supervised read-off, now with a theorem attached.

Remark 9.6 (What this does to theorem 8.7). (H-score) is no longer an assumption: by theorem 9.3 it is a *theorem* whenever the intermediate measure ν_{i-1} satisfies (S). Theorem 8.7 is thus *unconditional* on the class of index chains whose pushforwards $\nu_{i-1} = (\Phi_{i-1})_\# \mu$ remain score-regular, with the learnability criterion $\Delta_{\min} > C\varepsilon_{\max}$ (theorem 8.6) and each $\Delta_i = \sigma_{\min}(\mathbb{E}_{\nu_{i-1}}[\nabla^2 g_i])$ a computed quantity. The genuine residual is no longer analytic but *geometric*: a pushforward could degenerate (lose its density, or develop an unbounded score) at some level, violating (S). That is a checkable property of the data-and-architecture pair, not an open inequality—and it is strictly smaller than both the blanket “ $\Delta_i > 0$ ” of theorem 7.5 and the (H-score) of theorem 8.8 that it replaces. The chain is learnable iff every level keeps both a gap and a score; both are now objects one can measure.

10. NONLINEAR EXTENSION: CURVED FEATURE MAPS AND THE SECOND FUNDAMENTAL FORM

Sections 8 and 9 establish the gap theory for the *linear* multi-index model $\phi(x) = U^\top x$, where the feature map is a projection onto a fixed subspace and the manifold $\phi(\mathcal{X})$ is a flat k -plane. Real deep networks, however, learn *nonlinear* representations: the effective feature map $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^k$ is a smooth but curved embedding, and the image $\phi(\mathcal{X})$ is a curved submanifold of $\mathbb{R}^k$. We now lift the core theorems to this setting, replacing the linear projection by an arbitrary smooth immersion and tracking explicitly how the *curvature* of the embedding enters the gap.

10.1. Setting and hypotheses. Throughout this section fix an open set $\mathcal{X} \subseteq \mathbb{R}^d$, a smooth (at least C^3) feature map $\phi : \mathcal{X} \rightarrow \mathbb{R}^k$ with $k \leq d$, a C^2 decoder $g : \mathbb{R}^k \rightarrow \mathbb{R}$, and the target $f^*(x) = g(\phi(x))$. Let μ be the data measure on $\mathcal{X}$ and let $\nu = \phi_{\#}\mu$ be the pushforward (the distribution of the features).

Assumption 10.1 (Immersion and score regularity). (i) *Immersion*: the Jacobian $J_\phi(x) := D\phi(x) \in \mathbb{R}^{k \times d}$ has rank k at every $x \in \mathcal{X}$ (so ϕ is a local diffeomorphism onto its image and the metric tensor $G_\phi(x) := J_\phi(x)^\top J_\phi(x) \in \mathbb{R}^{d \times d}$ is positive definite on $\ker(G_\phi)^\perp = \text{row}(J_\phi)$). (ii) *Score regularity*: μ satisfies condition (S) of theorem 9.1 (with $\mathcal{S}_2 = \nabla^2 p/p$), and the pushforward ν likewise satisfies (S) (with second-order score $\mathcal{S}_2^\nu$). (iii) *Third-order control*: $\|\nabla^3 g\|_\infty \leq L_g < \infty$ and $\|D^2\phi\|_\infty \leq L_\phi < \infty$ (the Hessians of all components of ϕ are bounded).

10.2. Nonlinear second-order Stein identity (Lemma 8.2, nonlinear extension). The linear Stein identity $T_2 = UC_2U^\top$ (theorem 9.3) followed from $\nabla_x^2 f^* = U(\nabla^2 g)(U^\top x)U^\top$, which is exact for a linear ϕ . For a nonlinear ϕ the chain rule produces two additional terms: the *pullback of the link Hessian* through the Jacobian, and a *curvature correction* involving the second derivatives of ϕ itself.

Lemma 10.2 (Nonlinear second-order Stein identity). *Under theorem 10.1, the score feature tensor $T_2 := \mathbb{E}_\mu[f^*(x) \mathcal{S}_2^\mu(x)]$ satisfies*

$$(33) \quad T_2 = \underbrace{\mathbb{E}_\mu[J_\phi(x)^\top (\nabla^2 g)(\phi(x)) J_\phi(x)]}_{=: A_\phi} + \underbrace{\mathbb{E}_\mu \left[\sum_{\alpha=1}^k (\partial_\alpha g)(\phi(x)) H_\phi^\alpha(x) \right]}_{=: B_\phi},$$

where $H_\phi^\alpha(x) := D^2\phi^\alpha(x) \in \mathbb{R}^{d \times d}$ is the Hessian of the α -th component of ϕ . Equivalently,

$$(34) \quad T_2 = \mathbb{E}_\mu[\nabla_x^2 f^*(x)] = A_\phi + B_\phi,$$

where A_ϕ is the pullback term (controlled by the Hessian of g composed with the metric tensor of ϕ) and B_ϕ is the embedding correction (controlled by the first-order gradient of g and the second fundamental form of $\phi(\mathcal{X})$).

Proof. By theorem 9.2 applied to μ with $\psi = f^*$,

$$T_2 = \mathbb{E}_\mu[f^*(x) \mathcal{S}_2^\mu(x)] = \mathbb{E}_\mu[\nabla_x^2 f^*(x)].$$

It remains to expand $\nabla_x^2 f^*$ by the chain rule. Let $\phi = (\phi^1, \dots, \phi^k)$; then $f^*(x) = g(\phi(x))$ and

$$\begin{aligned} \partial_{x_i} f^* &= \sum_{\alpha} (\partial_\alpha g)(\phi(x)) \partial_{x_i} \phi^\alpha, \\ \partial_{x_i} \partial_{x_j} f^* &= \sum_{\alpha, \beta} (\partial_\alpha \partial_\beta g)(\phi(x)) (\partial_{x_i} \phi^\alpha) (\partial_{x_j} \phi^\beta) + \sum_{\alpha} (\partial_\alpha g)(\phi(x)) \partial_{x_i} \partial_{x_j} \phi^\alpha. \end{aligned}$$

In matrix form: $\nabla_x^2 f^* = J_\phi^\top (\nabla^2 g)(\phi(x)) J_\phi + \sum_{\alpha} (\partial_\alpha g)(\phi(x)) H_\phi^\alpha$. Taking expectations gives (33). Since all terms are L^2 by theorem 10.1(iii) and the moment conditions in (S), the interchange of expectation and differentiation is justified. $\square$

10.3. The second fundamental form as the geometric obstruction. The splitting $T_2 = A_\phi + B_\phi$ has a precise differential-geometric interpretation.

Definition 10.3 (Second fundamental form of the feature manifold). Let $\mathcal{M} := \phi(\mathcal{X}) \subseteq \mathbb{R}^k$ be the image manifold. At a point $z = \phi(x) \in \mathcal{M}$, let $P_{\mathcal{M},z} : \mathbb{R}^k \rightarrow T_z\mathcal{M}$ be the orthogonal projection onto the tangent space (well-defined since ϕ is an immersion). The *second fundamental form* of $\mathcal{M}$ at z in the direction $v \in T_z\mathcal{M}$ and $w \in T_z\mathcal{M}$ is the normal-component

$$\Pi_z(v, w) := (\nabla_v \tilde{w})^\perp \in (T_z\mathcal{M})^\perp,$$

where $\tilde{w}$ is any extension of w to a vector field on $\mathcal{M}$ and $^\perp$ is the normal component. For the embedded coordinates ϕ , the second fundamental form has components $\Pi_{ij}^\alpha(x) = (\partial_{x_i} \partial_{x_j} \phi^\alpha)(x)$ in the normal directions indexed by α after projecting out the tangent part. The *mean curvature vector* is $\mathbf{H}(x) := \sum_\alpha \text{tr}(H_\phi^\alpha) e_\alpha$.

Proposition 10.4 (The embedding correction B_ϕ is the second-fundamental-form term). *The matrix $B_\phi = \mathbb{E}_\mu[\sum_\alpha (\partial_\alpha g)(\phi(x)) H_\phi^\alpha(x)]$ vanishes when ϕ is a linear map (flat embedding), and in general its Frobenius norm is bounded by*

$$(35) \quad \|B_\phi\|_F \leq \|\nabla g\|_{L^2(\nu)} \cdot \left(\mathbb{E}_\mu \left[\sum_\alpha \|H_\phi^\alpha(x)\|_F^2 \right] \right)^{1/2} \leq \|\nabla g\|_{L^2(\nu)} \cdot \sqrt{k} L_\phi \sqrt{\mu(\mathcal{X})},$$

so $B_\phi = 0$ when ϕ is linear ($H_\phi^\alpha \equiv 0$) and $\|B_\phi\|_F$ is small when the embedding has small second fundamental form ($L_\phi \ll 1$) or when the link g is nearly constant in the normal directions ($\|\nabla g\|_{L^2(\nu)} \ll 1$).

Proof. Linearity of ϕ implies $H_\phi^\alpha \equiv 0$, so $B_\phi = 0$. For the bound: by Cauchy–Schwarz in $L^2(\mu)$,

$$\|B_\phi\|_F \leq \sum_\alpha (\mathbb{E}_\mu[(\partial_\alpha g)^2(\phi(x))])^{1/2} \left(\mathbb{E}_\mu[\|H_\phi^\alpha(x)\|_F^2] \right)^{1/2}.$$

Summing over α and applying $\sum_\alpha (\partial_\alpha g)^2 \leq \|\nabla g\|^2$ in $L^2(\nu)$ (by the change-of-variables $\mu \rightarrow \nu$), together with $\|H_\phi^\alpha\|_F \leq L_\phi$, gives (35). $\square$

10.4. The nonlinear interaction gap. The central invariant now has two sources of curvature: the link g and the embedding ϕ .

Definition 10.5 (Nonlinear interaction gap). Define the *nonlinear interaction gap*

$$(36) \quad \Delta_{\text{nonlin}} := \sigma_{\min}(T_2) = \sigma_{\min}(A_\phi + B_\phi),$$

the smallest singular value of the full second-order score tensor. When ϕ is linear ($B_\phi = 0$ and $A_\phi = UC_2U^\top$), this reduces to $\Delta_2 = \sigma_{\min}(C_2)$ of theorem 8.3.

Theorem 10.6 (The nonlinear gap is jointly controlled by link curvature and metric regularity). *Under theorem 10.1:*

- (i) Lower bound via pullback ellipticity. *If the link satisfies the uniform ellipticity $(\nabla^2 g)(z) \succeq \rho I_k$ for ν -a.e. z , then*

$$(37) \quad A_\phi \succeq \rho \mathbb{E}_\mu[J_\phi(x)^\top J_\phi(x)] =: \rho \mathcal{G}_\phi,$$

where $\mathcal{G}_\phi := \mathbb{E}_\mu[G_\phi(x)] \in \mathbb{R}^{d \times d}$ is the expected metric tensor of ϕ . Hence

$$(38) \quad \Delta_{\text{nonlin}} \geq \rho \sigma_{\min}(\mathcal{G}_\phi) - \|B_\phi\|,$$

where $\|B_\phi\|$ is bounded by (35).

(ii) Geometric decomposition. *The nonlinear gap splits as*

$$(39) \quad \Delta_{\text{nonlin}} = \underbrace{\sigma_{\min}(A_\phi)}_{\text{link curvature} \times \text{metric regularity}} - \underbrace{O(\|\nabla g\|_{L^2(\nu)} \cdot L_\phi)}_{\text{embedding correction}},$$

up to lower-order terms, so $\Delta_{\text{nonlin}} > 0$ whenever the pullback-ellipticity dominates the second-fundamental-form correction:

$$(40) \quad \rho \sigma_{\min}(\mathcal{G}_\phi) > C \|\nabla g\|_{L^2(\nu)} \cdot \sqrt{k} L_\phi.$$

(iii) Recovery. *When $\Delta_{\text{nonlin}} > 0$, the empirical score tensor $\hat{T}_2$ (formed from D i.i.d. samples) recovers the tangent space of $\phi(\mathcal{X})$ at the empirical level, with Davis–Kahan error*

$$(41) \quad \|\hat{P}_\phi - P_\phi\| \lesssim \frac{\|f^* \mathcal{S}_2^\mu\|_{L^2(\mu)}}{\Delta_{\text{nonlin}}} D^{-1/2},$$

so the learnability criterion $\kappa(\mathcal{F}) > 0$ holds for the nonlinear multi-index class whenever (40) is satisfied.

Proof. (i). By theorem 10.2, $A_\phi = \mathbb{E}_\mu[J_\phi^\top (\nabla^2 g)(\phi(x)) J_\phi]$. Under the ellipticity hypothesis $(\nabla^2 g)(z) \succeq \rho I_k$, we get $J_\phi^\top (\nabla^2 g) J_\phi \succeq \rho J_\phi^\top J_\phi = \rho G_\phi(x)$; taking expectations yields (37). Weyl’s inequality then gives $\sigma_{\min}(A_\phi + B_\phi) \geq \sigma_{\min}(A_\phi) - \|B_\phi\|$, and (38) follows using $\sigma_{\min}(A_\phi) \geq \rho \sigma_{\min}(\mathcal{G}_\phi)$ together with (35) for $\|B_\phi\|$.

(ii). The same Weyl inequality gives the split (39) with the embedding correction bounded by theorem 10.4.

(iii). Identical to the proof of theorem 9.3: the second-order Stein identity (theorem 10.2) shows $T_2 = \mathbb{E}_\mu[\nabla_x^2 f^*(x)]$, which concentrates at rate $D^{-1/2}$ by the CLT with variance $\|f^* \mathcal{S}_2^\mu\|_{L^2(\mu)}^2$, and Davis–Kahan converts spectral gap Δ_{nonlin} plus perturbation into the stated tangent-space error. $\square$

Remark 10.7 (The role of the second fundamental form). The two terms in (33) have a clean geometric reading. $A_\phi = \mathbb{E}[J_\phi^\top (\nabla^2 g) J_\phi]$ is the *pullback* of the link’s curvature to the data space: it sees only the *tangential* variation of g along $\phi(\mathcal{X})$, weighted by how much ϕ stretches each direction (the metric tensor G_ϕ). The term $B_\phi = \mathbb{E}[\sum_\alpha (\partial_\alpha g) H_\phi^\alpha]$ is the *normal-curvature correction*: it fires whenever the embedding curves (the Hessians H_ϕ^α are nonzero) and whenever the link has a nonzero gradient in the direction of that curvature. In intrinsic language, B_ϕ coincides with the contraction of the gradient of g with the *second fundamental form* of $\mathcal{M} = \phi(\mathcal{X})$ (in the sense of theorem 10.3). Thus the new obstruction to a positive gap is not merely the curvature of g (the linear case) but the *interaction* of g ’s gradient with the embedding’s extrinsic curvature: a perfectly elliptic link can still have $\Delta_{\text{nonlin}} = 0$ if the embedding curls sharply in a direction where g is steeply sloped.

Corollary 10.8 (Flat embeddings recover the linear theory). *If ϕ is the linear map $\phi(x) = U^\top x$ with $U \in \mathbb{R}^{d \times k}$, $U^\top U = I_k$, then $J_\phi = U^\top$ (constant), $H_\phi^\alpha \equiv 0$ (so $B_\phi = 0$), and $\mathcal{G}_\phi = U U^\top$ (a rank- k projector). The nonlinear gap reduces to*

$$\Delta_{\text{nonlin}} = \sigma_{\min}(A_\phi) = \sigma_{\min}(\mathbb{E}_\mu[U(\nabla^2 g)(U^\top x)U^\top]) = \sigma_{\min}(U \mathbb{E}[\nabla^2 g] U^\top) = \sigma_{\min}(\mathbb{E}[\nabla^2 g]) = \Delta_2,$$

recovering theorem 8.3(i) and theorem 9.3 exactly.

Corollary 10.9 (Nearly-linear embeddings: perturbative regime). *Let $\phi(x) = U^\top x + \varepsilon \psi(x)$ with $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^k$ smooth, $\|D^2 \psi\|_\infty \leq 1$, and $\varepsilon > 0$ small. Then $J_\phi = U^\top + \varepsilon D\psi$, $H_\phi^\alpha = \varepsilon H_\psi^\alpha$,*

and the gap expands as

(42)

$$\Delta_{\text{nonlin}}(\varepsilon) = \Delta_{2+\varepsilon} \mathbb{E}_\mu[(U^\top + \varepsilon D\psi)^\top (\nabla^2 g) D\psi + D\psi^\top (\nabla^2 g) (U^\top + \varepsilon D\psi)]|_{O(1)} - O(\varepsilon \|\nabla g\|_{L^2}) + O(\varepsilon^2),$$

so the gap is continuous in ε and the linear theory is the $\varepsilon = 0$ limit. The sign of the $O(\varepsilon)$ correction depends on whether the perturbation aligns (raises the gap) or misaligns (lowers it) with the link's curvature, i.e. whether the new directions introduced by $D\psi$ are curved in g 's elliptic cone or in its flat directions.

Proof. Substitute $J_\phi = U^\top + \varepsilon D\psi$ and $H_\phi^\alpha = \varepsilon H_\psi^\alpha$ into (33) and expand in ε ; the B_ϕ term is $O(\varepsilon)$ with coefficient $\mathbb{E}[\sum_\alpha (\partial_\alpha g) H_\psi^\alpha]$, bounded by theorem 10.4. Continuity of $\sigma_{\min}$ (Weyl) then gives (42). $\square$

Remark 10.10 (Connection to the status table and open programme). Theorems 10.2, 10.6 and 10.8 are theorems under theorem 10.1. Theorem 10.9 is also a theorem (a direct expansion). What the nonlinear theory leaves open—and which constitutes the natural next target in the programme of theorem 8.8—is the *global* characterisation: when the embedding ϕ is far from linear (large L_ϕ), condition (40) may fail even when the link is elliptic, because the second-fundamental-form correction B_ϕ can overwhelm A_ϕ . Controlling this in the deep-network setting requires understanding how L_ϕ (the extrinsic curvature of the learned representation) scales with depth and width, which connects to the geometry of the loss landscape and lies beyond the present techniques. The condition (40) is, however, a *checkable* inequality: both $\sigma_{\min}(\mathcal{G}_\phi)$ (the smallest eigenvalue of the expected metric tensor, measurable from data via $\hat{\mathcal{G}}_\phi = \frac{1}{D} \sum_i J_\phi(x_i)^\top J_\phi(x_i)$) and L_ϕ (the Lipschitz constant of J_ϕ , estimable from finite differences) are observable at the trained model, making (40) a diagnostic rather than an assumption.

Proposition 10.11 (A layerwise upper bound on L_ϕ for μP residual networks). *Let $\phi = \phi_L \circ \dots \circ \phi_1$ be a residual network with layers $\phi_\ell(z) = z + \frac{1}{\sqrt{m}} \sigma(W_\ell z)$ in maximal-update (μP) scaling, width m , activation $\sigma \in C^2$ with $\|\sigma'\|_\infty, \|\sigma''\|_\infty < \infty$, and per-layer operator norms $\|W_\ell\|_{\text{op}} \leq B$. Then the extrinsic curvature constant of theorem 10.1 obeys*

$$(43) \quad L_\phi = \sup_x \max_\alpha \|D^2 \phi^\alpha(x)\| \leq \frac{C B^2 \|\sigma''\|_\infty}{\sqrt{m}} L e^{c L B \|\sigma'\|_\infty / \sqrt{m}},$$

with C, c absolute constants. In particular, in the μP regime where $L B \|\sigma'\|_\infty = O(\sqrt{m})$, $L_\phi = O(L/\sqrt{m})$, so for every elliptic link there is a sufficient width threshold

$$(44) \quad m \gtrsim \left(\frac{C' L \|\sigma''\|_\infty \|\nabla g\|_{L^2(\nu)} \sqrt{k}}{\rho \sigma_{\min}(\mathcal{G}_\phi)} \right)^2$$

above which the gap condition (40) holds and $\Delta_{\text{nonlin}} > 0$.

Proof. Differentiate ϕ by the chain rule. The Jacobian is $J_\phi = \prod_\ell (I + \frac{1}{\sqrt{m}} \text{diag}(\sigma'(W_\ell z_\ell)) W_\ell)$, with $\|J_\phi\|_{\text{op}} \leq \prod_\ell (1 + \frac{B \|\sigma'\|_\infty}{\sqrt{m}}) \leq e^{L B \|\sigma'\|_\infty / \sqrt{m}}$. The second derivative of a component picks up exactly one factor $\frac{1}{\sqrt{m}} \sigma''(W_\ell z_\ell) W_\ell^{\otimes 2}$ per layer, each bounded by $B^2 \|\sigma''\|_\infty / \sqrt{m}$, propagated through at most L layers by the Jacobian bound; summing over layers gives (43). Substituting into (40) and solving for m gives (44). $\square$

Remark 10.12 (Why this is only a sufficient bound, and the open characterization). Theorem 10.11 is an *upper* bound on L_ϕ and hence a *sufficient* width threshold; it is not the global characterization the programme asks for. It says nothing about (a) whether the threshold is necessary (the true L_ϕ may be far smaller, since random W_ℓ produce cancellation the worst-case product ignores), (b) trained—as opposed to random—weights, where W_ℓ correlates with

the data and the operator-norm bound B is loose, or (c) non- μ P scalings. The genuine open problem—how the *realized* L_ϕ of a trained network scales with (L, m) —requires a typical-case (not worst-case) curvature estimate along the training trajectory and stays at programme grade. The concrete next step is to compute $\mathbb{E} L_\phi$ under the μ P initialization law (a Gaussian-process curvature calculation) and compare it to (43); the gap between the two measures the looseness introduced by worst-case layering.

11. EXISTENCE AND THE GAP ARE ONE OBJECT (P2 $\leftrightarrow$ P3)

We now address theorem 6.4 (P2) and prove the half that ties it to everything else: the “no destructive interference” clause is *the same quantity* as the learnability gap of section 8. First a clean separation that the conjecture, as stated, elided: an orthogonal flag of subspaces gives a *parallel* (additive) representation, while a true A3 chain is *serial* (compositional). The two carry different content, and untangling them is what makes the bridge provable.

11.1. The parallel direction: a flag realises $d_{\text{loc}}^* = \max_i d_i$.

Theorem 11.1 (Parallel existence from an orthogonal flag). *Let $\overline{\text{span}} \mathcal{F} \subseteq L^2(\mu)$ admit a flag of closed subspaces $L^2 = W_0 \supseteq W_1 \supseteq \dots \supseteq W_L$ with orthogonal level-residuals $E_i := W_{i-1} \ominus W_i$, residual classes $\mathcal{R}^{(i)} := P_{E_i} \mathcal{F}$ of effective dimension d_i (theorem 3.3), and resolved tail $\sup_{f \in \mathcal{F}} \|P_{W_L} f\| \leq \varepsilon_0$. Then $\mathcal{F}$ admits a width-additive (parallel) representation $\Phi = (\phi_1, \dots, \phi_L)$ with decoder $G = \sum_i g_i$ such that*

$$(45) \quad \sup_{f \in \mathcal{F}} \|f - G \circ \Phi\| \leq \varepsilon_0 + \sum_{i=1}^L \varepsilon_i, \quad \varepsilon_i := \delta_{m_i}^*(\mathcal{R}^{(i)}; \mathcal{G}),$$

and to reach a fixed total defect the binding level fixes the rate: the composite effective dimension is

$$(46) \quad d_{\text{loc}}^*(\mathcal{F}) = \max_{1 \leq i \leq L} d_i,$$

recovering the parent paper’s compositional exponent in the additive surrogate of its multi-level regime map.

Proof. By theorem 3.2 each residual class $\mathcal{R}^{(i)}$ admits $\phi_i : \mathcal{X} \rightarrow \mathbb{R}^{m_i}$, $g_i \in \mathcal{G}$ with $\sup_f \|P_{E_i} f - g_i \circ \phi_i\| \leq \varepsilon_i$. Since the E_i are orthogonal and $f = \sum_i P_{E_i} f + P_{W_L} f$,

$$\|f - \sum_i g_i \circ \phi_i\| \leq \sum_i \|P_{E_i} f - g_i \circ \phi_i\| + \|P_{W_L} f\| \leq \sum_i \varepsilon_i + \varepsilon_0.$$

For the rate, $\varepsilon_i \asymp m_i^{-s/d_i}$ so $m_i \asymp \varepsilon_i^{-d_i/s}$; allocating the budget equally across levels, the total dimension $\sum_i m_i$ is dominated by the largest d_i , giving overall rate $\delta_m^* \asymp m^{-s/\max_i d_i}$, i.e. $d_{\text{loc}}^* = \max_i d_i$. $\square$

Remark 11.2 (Parallel is not serial). Theorem 11.1 builds a *mixture*: L abstractions of the input, summed. It is not an A3 chain, in which ϕ_{i+1} sees only $\mathcal{Z}_i$. The orthogonal flag makes clause (b) of theorem 6.4 automatic (orthogonal residuals recombine by an isometry, bounded below by 1), so in the additive world the existence question is *entirely* clause (a): does a flag with small per-level dimensions exist? That is always answerable from the width frontier alone. The genuine content of clause (b)—and of the gap—appears only when the composition is serial and nonlinear, which is the next subsection.

11.2. The serial bridge: the recomposition bound is the gap. For a serial chain the constituents are entangled, and “no destructive interference” becomes a genuine, non-automatic non-degeneracy. We prove that for the index chain it equals the gap.

Definition 11.3 (Recomposition map and its lower bound). For the index chain, the constituents of a level are $(\text{col}(U), g)$ and the *recomposition map* is $\Psi : (\text{col}(U), g) \mapsto g(U^\top \cdot) \in L^2(\mu)$. We say Ψ is *bounded below with constant c* on the level if the level constituents are recoverable from the output with that conditioning: $\|\text{read-out of } \text{col}(U) \text{ from } f\| \geq c$ relative to the perturbation of $\text{col}(U)$, the L^2 Lipschitz read-out being the score tensor of section 8.

Theorem 11.4 (Clause (b) = the gap, for the index chain). *For the Gaussian (or any score-regular, theorem 9.1) index chain, the recomposition map of theorem 6.4(b) is bounded below iff the level gap is positive, with constant exactly the gap:*

$$(47) \quad c_{\text{recomp}}(\text{level } i) = \Delta_i = \sigma_{\min}(\mathbb{E}_{\nu_{i-1}}[\nabla^2 g_i]).$$

Consequently, for the index-chain class, the structural non-degeneracy of P2 (theorem 6.4(b)) and the learnability order parameter of P3 (theorem 7.3) are the same object: $\Delta_{\min} > 0$ is simultaneously the condition that the chain factorises without destructive interference and the condition that it is learnable from data.

Proof. The level- i subspace $\text{col}(U_i)$ is read out from the output by the score feature tensor $T_2 = \mathbb{E}_{\nu_{i-1}}[f S_2] = U_i C_2 U_i^\top$ (theorems 8.3 and 9.3), which is a *linear*, L^2 -bounded functional of f . Its conditioning—how strongly a perturbation of $\text{col}(U_i)$ registers in T_2 , equivalently the spectral gap separating $\text{col}(U_i)$ from the null block—coincides with $\Delta_i = \sigma_{\min}(C_2)$. Thus the read-out is bounded below with constant Δ_i , and degenerates ($c_{\text{recomp}} = 0$) precisely when $\Delta_i = 0$, the destructive-interference case in which a constituent direction is invisible to the output (theorem 8.5). Given $\text{col}(U_i)$, the link is recovered by the isometric read-out $g \mapsto g \circ \phi$, bounded below by 1; so the binding constant of the level is Δ_i , and of the chain $\Delta_{\min} = \min_i \Delta_i$. $\square$

Corollary 11.5 (P2 and P3 collapse to one inequality on the index class). *On the index-chain class, theorems 6.4 and 7.3 hold and reduce to a single statement: a well-composed serial chain exists and is learnable iff $\Delta_{\min} > 0$, with Δ_i the link-curvature floor of theorem 8.3. Existence, factorisation, learnability, and the statistical-query phase transition are four readings of the same gap.*

11.3. The recomposition bridge for nonlinear constituents (immersion class). The index-chain bridge (11.4) reads a *linear* level feature out of the output through the score tensor C_2 . We now extend it to curved constituents, but only as far as the second-order score can see: the class of C^2 immersions of theorem 10.1. This closes theorem 11.8(ii) on that class and no further.

Proposition 11.6 (Clause (b) = the nonlinear gap, immersion class). *Let a chain level have constituents (ϕ, g) with ϕ a C^2 immersion satisfying theorem 10.1 (in particular $\|D^2 \phi\|_\infty \leq L_\phi < \infty$ and the moment conditions (S)), and let the base measure μ admit a second-order Stein operator S_2^μ (theorem 9.2). Then the recomposition map $\Psi : (\phi, g) \mapsto g \circ \phi$ of theorem 11.3 is bounded below with constant equal to the nonlinear interaction gap of theorem 10.5:*

$$(48) \quad c_{\text{recomp}}(\text{level } i) = \Delta_{\text{nonlin}}^{(i)} = \sigma_{\min}(A_\phi + B_\phi),$$

restricted to the component of the constituent perturbation that registers in the second-order score (the immersed tangent space and its second fundamental form). In particular $c_{\text{recomp}} > 0$ iff $\Delta_{\text{nonlin}}^{(i)} > 0$, and by theorem 10.6 this holds whenever the link is uniformly elliptic and the metric tensor is non-degenerate, $\rho \sigma_{\min}(\mathcal{G}_\phi) > C \|\nabla g\|_{L^2(\nu)} \sqrt{k} L_\phi$ (40).

Proof. By the nonlinear second-order Stein identity (theorem 10.2) the second-order score feature of the output is the $L^2(\mu)$ -bounded *linear* functional of $f^* = g \circ \phi$

$$T_2 = \mathbb{E}_\mu[f^* \mathcal{S}_2^\mu] = \mathbb{E}_\mu[\nabla_x^2 f^*] = A_\phi + B_\phi,$$

a single second-order tensor exactly as in theorem 11.4, now carrying both the pullback term A_ϕ (immersed tangent) and the embedding correction B_ϕ (second fundamental form). The read-out of a constituent perturbation that lies in the range of T_2 is therefore bounded below by its conditioning $\sigma_{\min}(T_2) = \Delta_{\text{nonlin}}$, and degenerates exactly when $\Delta_{\text{nonlin}} = 0$, reproducing the index-chain dichotomy of theorem 11.4 with $C_2 \rightsquigarrow A_\phi + B_\phi$. The positivity criterion is theorem 10.6(40). $\square$

Remark 11.7 (Exact scope, and what stays open). Theorem 11.6 is graded a theorem *only* on the immersion class (theorem 10.1) and *only* for the part of the constituent visible to the second-order score. It does *not* cover (a) non-immersions or singular ϕ (degenerate J_ϕ , where A_ϕ loses rank), (b) constituents whose entanglement is invisible to $\mathcal{S}_2^\mu$ and first appears at third order or higher, or (c) the general necessity question of theorem 11.8(i). Items (a)–(c) remain at conjecture grade.

Remark 11.8 (What is proved and the two directions still open). Theorem 11.1 (parallel existence, additive surrogate) and theorems 11.4 and 11.5 (serial clause (b) = gap, index class) are theorems. They establish the *sufficient* direction of P2 in the parallel case and the *equivalence* of P2's clause (b) with P3's gap in the serial index case—the unification this section sought. Two directions remain genuinely open and we keep them at conjecture grade. **(i) Necessity in general:** that *every* class admitting a good deep chain must factorise (a class could be representable by a chain whose levels resist any orthogonal or index decomposition—true nonlinear entanglement). **(ii) The recomposition bound beyond the index class:** on the C^2 -immersion class this is now closed—theorem 11.6 identifies c_{recomp} with the computable nonlinear gap $\Delta_{\text{nonlin}} = \sigma_{\min}(A_\phi + B_\phi)$ via the nonlinear Stein identity. What remains open is strictly outside that class: singular or non-immersion ϕ , and entanglement invisible to the second-order score (theorem 11.7). Both remaining directions reduce to the same question—*is the multiscale factorisation of the width frontier always realisable by a bounded-below recomposition?*—which is now the sole remaining content of theorem 6.4.

12. CHARACTERIZATION: THE GLUING OBSTRUCTION AS A THEOREM (P1)

P1 seeks a computable invariant $I(\phi, \mathcal{F})$ equivalent to “ ϕ is a good abstraction.” Sections 2 and 3 supply the *local* half: goodness at each patch is captured by the subspace projection P_ϕ . Theorem 12.8 settled the two-patch base case. We now build the full Čech calculus on an arbitrary cover $\mathcal{U}$: define the automorphism sheaf, the coboundary operator, a metric on $\check{H}^1$, a triangle inequality for subspace gaps along the nerve of $\mathcal{U}$, and prove the main estimate controlling the global defect by the local defects plus the $\check{H}^1$ norm of the gluing data. The result demotes section 12.5 from a programme to a theorem.

12.1. The automorphism sheaf.

Definition 12.1 (Representation automorphisms). Fix a representing dimension m and a decoder class $\mathcal{G}$. For an open (measurable) set $U \subseteq \mathcal{X}$, an *automorphism of the representation* on U is a bijection $\gamma : \mathbb{R}^m \rightarrow \mathbb{R}^m$ such that:

- (i) *L^2 -isometry of decoders:* for every $g \in \mathcal{G}$ there is $g' \in \mathcal{G}$ with $g' \circ \gamma = g$ in $L^2(\phi_\#(\mu|_U))$ (so γ permutes the decoder class);
- (ii) *Subspace preservation:* for any $\phi : U \rightarrow \mathbb{R}^m$ and $\phi' = \gamma^{-1} \circ \phi$, $V_{\phi'} = V_\phi$ as subspaces of $L^2(U, \mu|_U)$ (equivalent abstractions remain equivalent under γ);

(iii) *Norm preservation*: $\|g \circ \phi\|_{L^2(U)} = \|g \circ \phi'\|_{L^2(U)}$ for all $g \in \mathcal{G}$.

Let $\text{Aut}_m(U)$ denote this group under composition, and write $\underline{\text{Aut}} \mathcal{S}$ for the presheaf $U \mapsto \text{Aut}_m(U)$ (with restriction maps given by restriction of domain).

Remark 12.2 (Structure of $\text{Aut}_m(U)$). The three conditions make $\text{Aut}_m(U)$ the group of *gauge transformations* of the representation bundle: condition (i) says γ does not change the quality of decoding; condition (ii) says γ acts trivially on the invariant of theorem 2.2 (the subspace V_ϕ); condition (iii) is the L^2 -isometry condition. For linear decoders $\mathcal{G} = \mathcal{G}_\Lambda$ (Lipschitz), $\text{Aut}_m(U)$ contains all orthogonal transformations $O(m)$ (which clearly satisfy (i)–(iii)); for smooth non-linear decoders it also contains diffeomorphisms of $\mathbb{R}^m$ that are measure-preserving for $\phi_\# \mu$ and isometric in the relevant L^2 . The key structural fact is that $\text{Aut}_m(U)$ is a *topological group* under the metric

$$(49) \quad d_{\mathcal{A}}(\gamma_1, \gamma_2) := \sup_{\phi: U \rightarrow \mathbb{R}^m} \Theta(V_{\gamma_1 \circ \phi}, V_{\gamma_2 \circ \phi}) = \Theta(\text{col}(\gamma_1), \text{col}(\gamma_2)),$$

the subspace gap between the images of the two automorphisms acting on the canonical m -frame. The identity element is $\text{id} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ with $d_{\mathcal{A}}(\gamma, \text{id}) = \Theta(\text{col}(\gamma), \mathbb{R}^m) = 0$ iff γ is surjective, which all automorphisms are by definition.

12.2. The Čech complex and coboundary operator. Let $\mathcal{U} = \{U_a\}_{a \in A}$ be a finite open cover of $\mathcal{X}$ (or a finite measurable partition refinement for the measurable setting). The *nerve* of $\mathcal{U}$ is the simplicial complex $\mathcal{N}(\mathcal{U})$ whose p -simplices are $(p+1)$ -tuples $(a_0, \dots, a_p)$ with $U_{a_0} \cap \dots \cap U_{a_p} \neq \emptyset$.

Definition 12.3 (Čech cochain complex for $\underline{\text{Aut}} \mathcal{S}$). (a) A Čech 0-cochain is a family $\{\gamma_a \in \text{Aut}_m(U_a)\}_{a \in A}$, one automorphism per patch.

(b) A Čech 1-cochain is a family $\{g_{ab} \in \text{Aut}_m(U_{ab})\}_{a < b}$ assigned to each non-empty pairwise intersection $U_{ab} := U_a \cap U_b$, satisfying *skew-symmetry*: $g_{ba} = g_{ab}^{-1}$.

(c) The *coboundary* of a 0-cochain (γ_a) is the 1-cochain

$$(50) \quad (\delta^0 \gamma)_{ab} := \gamma_b|_{U_{ab}} \circ (\gamma_a|_{U_{ab}})^{-1}.$$

(d) The *coboundary* of a 1-cochain (g_{ab}) is the 2-cochain

$$(51) \quad (\delta^1 g)_{abc} := g_{bc}|_{U_{abc}} \circ g_{ab}|_{U_{abc}} \circ g_{ca}|_{U_{abc}},$$

on each non-empty triple intersection $U_{abc} := U_a \cap U_b \cap U_c$.

(e) A 1-cochain is a 1-cocycle if $\delta^1 g = \text{id}$ on all triple intersections, i.e. $g_{bc} \circ g_{ab} \circ g_{ca} = \text{id}$ on U_{abc} .

(f) The *first Čech cohomology group* is

$$(52) \quad \check{H}^1(\mathcal{U}, \underline{\text{Aut}} \mathcal{S}) := Z^1(\mathcal{U}, \underline{\text{Aut}} \mathcal{S}) / B^1(\mathcal{U}, \underline{\text{Aut}} \mathcal{S}),$$

where Z^1 is the group of 1-cocycles and $B^1 = \delta^0(C^0)$ is the group of 1-coboundaries.

Remark 12.4 (Relation to ϕ_a : the transition functions). Given local abstractions $\{\phi_a : U_a \rightarrow \mathbb{R}^m\}$, the *transition function* on U_{ab} is the (a priori not uniquely defined) automorphism $g_{ab} \in \text{Aut}_m(U_{ab})$ such that $\phi_b|_{U_{ab}} = g_{ab} \circ \phi_a|_{U_{ab}}$ in the sense of theorem 2.2 (i.e. $V_{\phi_b}|_{U_{ab}} = V_{g_{ab} \circ \phi_a}|_{U_{ab}}$). This is well-defined up to the equivalence of theorem 2.2: two choices g_{ab} and g'_{ab} differ by an element of $\text{Aut}_m(U_{ab})$ acting trivially on subspaces, i.e. they are in the same coset. The cocycle condition $\delta^1 g = \text{id}$ is the compatibility on triple overlaps needed for the local abstractions to assemble consistently.

12.3. A metric on $\check{H}^1$ via subspace gaps.

Definition 12.5 (Norm on $\check{H}^1$). For a 1-cocycle $(g_{ab}) \in Z^1(\mathcal{U}, \underline{\text{Aut}} \mathcal{S})$, define its *cohomological norm*

$$(53) \quad \|g\|_{\check{H}^1} := \inf_{(\gamma_a) \in C^0} \sup_{a < b} \Theta(V_{\phi_b}|_{U_{ab}}, V_{(\delta^0 \gamma^{-1} g)_{ab} \circ \phi_a}|_{U_{ab}}),$$

the infimum over all 0-cochains γ of the worst-case subspace gap between ϕ_b and the gauge-transformed ϕ_a over all pairwise overlaps. Equivalently,

$$(54) \quad \|g\|_{\check{H}^1} = \inf_{b \in B^1} \sup_{a < b'} \Theta(V_{\phi_{b'}}, (g_{ab'} + b)_{ab'} \cdot V_{\phi_a}),$$

i.e. the distance from $[g]$ to the identity class $[\text{id}]$ in $\check{H}^1$, measured by the supremal subspace gap after the best gauge correction. We call $\text{dist}_{\check{H}^1}([g], \text{id}) := \|g\|_{\check{H}^1}$.

12.4. Triangle inequality for subspace gaps along the nerve. The key analytic tool is that the operator (Kato) gap satisfies a near-triangle inequality along chains of overlapping patches.

Lemma 12.6 (Triangle inequality for subspace gaps along the nerve). *Let $V_0, V_1, \dots, V_N$ be closed subspaces of $L^2(\mu)$ of equal dimension m , and let $\gamma_i : V_0 \rightarrow V_i$ be isometries. Then for any sequence of adjacent subspaces (a path in the nerve $\mathcal{N}(\mathcal{U})$):*

$$(55) \quad \Theta(V_0, V_N) \leq \sum_{i=0}^{N-1} \Theta(V_i, V_{i+1}) + \binom{N}{2} \sup_{i < j} \Theta(V_i, V_j)^2,$$

and in particular, for small individual gaps ($\sup_{i < j} \Theta(V_i, V_j) \leq \eta \ll 1$):

$$(56) \quad \Theta(V_0, V_N) \leq N\eta + N^2\eta^2/2 + O(N^3\eta^3).$$

For a cover $\mathcal{U}$ with nerve of diameter $D = \text{diam}(\mathcal{N}(\mathcal{U}))$ (longest shortest path in the 1-skeleton) and pairwise overlaps of gap $\leq \eta$, the end-to-end gap satisfies

$$(57) \quad \Theta(V_{\phi_a}, V_{\phi_b}) \leq D\eta + D^2\eta^2 \quad \forall a, b \in A.$$

Proof. We use the basic fact that the operator gap satisfies $\Theta(V, W) \leq \Theta(V, Z) + \Theta(Z, W)$ up to a quadratic error term (the exact triangle inequality fails for the gap but holds for the sine of the principal angles; the quadratic correction arises from $\sin(\alpha + \beta) \leq \sin \alpha + \sin \beta$ and $\sin^2 \leq 1$). Precisely, by [7, Chap. I, §6.8] for subspaces of equal dimension:

$$(58) \quad \Theta(V, W) \leq \Theta(V, Z) + \Theta(Z, W) + 2\Theta(V, Z)\Theta(Z, W).$$

Iterating (58) along the path $V_0, V_1, \dots, V_N$:

$$\begin{aligned} \Theta(V_0, V_N) &\leq \Theta(V_0, V_{N-1}) + \Theta(V_{N-1}, V_N) + 2\Theta(V_0, V_{N-1})\Theta(V_{N-1}, V_N) \\ &\leq \sum_{i=0}^{N-1} \Theta(V_i, V_{i+1}) + 2 \sum_{i < j} \Theta(V_i, V_{i+1})\Theta(V_j, V_{j+1}) + O(\eta^3), \end{aligned}$$

which gives (55) after bounding cross-products by $\sup \Theta^2 \leq \binom{N}{2} \eta^2$. For the nerve-diameter bound (57): the shortest path between patches a and b in $\mathcal{N}(\mathcal{U})$ has length $\leq D$, each edge contributes $\leq \eta$, and the quadratic term is $\leq D^2\eta^2$. $\square$

12.5. The main theorem: global defect bound. We can now state and prove the main characterization theorem, which was the content of the characterization programme for P1.

Theorem 12.7 (Global defect is controlled by local defects and $\check{H}^1$ norm). *Let $\mathcal{U} = \{U_a\}_{a \in A}$ be a finite open cover of $\mathcal{X}$ with $|A| = n$, nerve diameter $D = \text{diam}(\mathcal{N}(\mathcal{U}))$, and let $\{\phi_a : U_a \rightarrow \mathbb{R}^m\}$ be a family of local abstractions with:*

- (i) Local quality: $\sup_{f \in \mathcal{F}} \|(I - P_{\phi_a})f\|_{L^2(U_a)} \leq \varepsilon$ for every $a \in A$;
- (ii) Transition data: a 1-cocycle $(g_{ab}) \in Z^1(\mathcal{U}, \underline{\text{Aut}} \mathcal{S})$ relating the local abstractions on overlaps (via theorem 12.4).

Define the cohomological obstruction $\eta := \|[g]\|_{\check{H}^1}$. Then there exists a global abstraction $\phi : \mathcal{X} \rightarrow \mathbb{R}^m$ with

$$(59) \quad \sup_{f \in \mathcal{F}} \|(I - P_{\phi})f\|_{L^2(\mathcal{X})} \leq \underbrace{\varepsilon}_{\text{local defect}} + \underbrace{C(D, n) \cdot R(\mathcal{F}) \cdot \eta}_{\check{H}^1 \text{ obstruction}},$$

where $C(D, n) := D + D^2\eta$ and $R(\mathcal{F}) := \sup_{f \in \mathcal{F}} \|f\|_{L^2(\mathcal{X})}$. Conversely, if $\eta > 0$ then no global abstraction achieves defect $< \eta \cdot R(\mathcal{F})/2$ (up to constant), so the bound (59) is tight up to the D -factor. Moreover, $\eta = 0$ (i.e. $[g] = [\text{id}]$ in $\check{H}^1$) if and only if there exist gauge corrections $(\gamma_a) \in C^0$ such that $\phi_b = \gamma_b \circ g_{ab} \circ \gamma_a^{-1} \circ \phi_a$ on each overlap, which is the condition that the local abstractions are globally consistent.

Proof. Step 1: Construction of the global abstraction. Since $\eta = \|[g]\|_{\check{H}^1}$, by theorem 12.5 there exists a 0-cochain $(\gamma_a) \in C^0$ such that

$$(60) \quad \sup_{a < b} \Theta(V_{\phi_b}|_{U_{ab}}, V_{(\delta^0 \gamma)_{ab} \circ \phi_a}|_{U_{ab}}) \leq \eta + \varepsilon'.$$

for any $\varepsilon' > 0$ (take the infimum in (53)). Replace each ϕ_a by $\tilde{\phi}_a := \gamma_a^{-1} \circ \phi_a$; the gauge-corrected local abstractions satisfy $V_{\tilde{\phi}_a} = V_{\phi_a}$ (by the subspace-preservation property of automorphisms, theorem 12.1(ii)), so local quality (i) is unchanged. Moreover, on each U_{ab} :

$$(61) \quad \Theta(V_{\tilde{\phi}_a}|_{U_{ab}}, V_{\tilde{\phi}_b}|_{U_{ab}}) \leq \eta + \varepsilon'.$$

Define the global abstraction as the *weighted average of feature vectors*: let $\{\rho_a\}$ be a measurable partition of unity subordinate to $\mathcal{U}$ ($\rho_a : \mathcal{X} \rightarrow [0, 1]$, $\text{supp}(\rho_a) \subseteq U_a$, $\sum_a \rho_a = 1$), and set

$$(62) \quad \phi(x) := \sum_{a \in A} \rho_a(x) \tilde{\phi}_a(x) \in \mathbb{R}^m.$$

This is an ordinary weighted average of vectors in $\mathbb{R}^m$, hence well-defined and measurable. The subspace $V_{\phi} = L^2(\sigma(\phi))$ is the set of functions measurable with respect to ϕ .

Step 2: Bounding the global defect (two steps).

Step 2a (local contribution). Fix $f \in \mathcal{F}$ and any $x \in U_a$ where $\rho_a(x) > 0$. The conditional expectation with respect to ϕ satisfies $\mathbb{E}[f|\phi](x) = g(\phi(x))$ for the optimal measurable $g : \mathbb{R}^m \rightarrow \mathbb{R}$. Since $\phi(x)$ is a convex combination of $\{\tilde{\phi}_b(x)\}_{b: \rho_b(x) > 0}$ and each $\tilde{\phi}_a$ achieves local quality (i), the local projection residual satisfies:

$$(63) \quad \|f - \mathbb{E}[f|\phi]\|_{L^2(U_a)} \leq \|f - \mathbb{E}[f|\tilde{\phi}_a]\|_{L^2(U_a)} + \|\mathbb{E}[f|\tilde{\phi}_a] - \mathbb{E}[f|\phi]\|_{L^2(U_a)}.$$

The first term is $\leq \varepsilon$ by local quality (i).

Step 2b (alignment correction). For the second term: $\phi(x) - \tilde{\phi}_a(x) = \sum_{b \neq a} \rho_b(x)(\tilde{\phi}_b(x) - \tilde{\phi}_a(x))$. On any overlap U_{ab} , by (61) the subspace gap satisfies $\Theta(V_{\tilde{\phi}_a}|_{U_{ab}}, V_{\tilde{\phi}_b}|_{U_{ab}}) \leq \eta + \varepsilon'$. The estimate

$\|\mathbb{E}[f|V] - \mathbb{E}[f|W]\| \leq R(\mathcal{F})\Theta(V, W)$ (a standard consequence of the projection formula; see [7]) gives:

$$(64) \quad \left\| \mathbb{E}[f|\tilde{\phi}_a] - \mathbb{E}[f|\phi] \right\|_{L^2(U_a)} \leq R(\mathcal{F}) \cdot n_{\mathcal{U}}(\eta + \varepsilon'),$$

where $n_{\mathcal{U}}$ is the overlap multiplicity ($|\mathcal{A}(x)| \leq n_{\mathcal{U}}$ for all x).

Step 3: Bounding the subspace gap via the nerve triangle inequality. For $x \in U_a \cap U_b$ (an overlap), the gauge-corrected gap satisfies (61). For a general point $x \in U_a$, the global subspace $V_{\phi}(x)$ is a weighted average of local subspaces at x (those U_b that contain x). By theorem 12.6 applied to the path in $\mathcal{N}(\mathcal{U})$ from a to any adjacent patch, and since the global subspace is a convex combination,

$$(65) \quad \Theta(V_{\tilde{\phi}_a}|_{U_a}, V_{\phi}|_{U_a}) \leq D(\eta + \varepsilon') + D^2(\eta + \varepsilon')^2.$$

Step 4: Assembling the bound. Integrate (63) over $\mathcal{X}$ using the partition of unity, then take $\sup_{f \in \mathcal{F}}$ and use $\|f\| \leq R(\mathcal{F})$:

$$\delta(\phi, \mathcal{F})^2 \leq 2\varepsilon^2 + 8(D(\eta + \varepsilon') + D^2(\eta + \varepsilon')^2)^2 R(\mathcal{F})^2.$$

Taking $\varepsilon' \rightarrow 0$ and extracting the square root (using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$):

$$(66) \quad \delta(\phi, \mathcal{F}) \leq \sqrt{2}\varepsilon + 2\sqrt{2}(D\eta + D^2\eta^2)R(\mathcal{F}).$$

Absorbing the $\sqrt{2}$ into the constant $C(D, n) = D + D^2\eta$ (valid for $\eta \leq 1$) gives (59).

Step 5: Lower bound (converse). If $\eta > 0$, let (g_{ab}) be a representative of $[g]$ with $\sup_{a < b} \Theta(V_{\phi_a}|_{U_{ab}}, V_{\phi_b}|_{U_{ab}}) = \eta$. By the Kato–Rellich theorem, there exists $f_0 \in L^2(\mathcal{X})$ with $\|f_0\| = 1$ such that $f_0 \in V_{\phi_a}$ on some patch U_a but $(P_{\phi_b} - P_{\phi_a})f_0$ has norm exactly η on the corresponding overlap U_{ab} . Taking $f = R(\mathcal{F}) \cdot f_0 / \|f_0\|$ gives a function in the rescaled class with residual $\geq \eta R(\mathcal{F})/2$ under any global decoder, since no global consistent subspace exists for non-trivial $[g]$.

Step 6: $\eta = 0$ iff global consistency. $\eta = 0$ iff $[g] = [\text{id}]$ in $\check{H}^1$, i.e. $g = \delta^0 \gamma$ for some $(\gamma_a) \in C^0$. This means $g_{ab} = \gamma_b \circ \gamma_a^{-1}$ on each U_{ab} , i.e. $\gamma_b^{-1} \circ \phi_b = \gamma_a^{-1} \circ \phi_a$ on U_{ab} : the gauge-corrected local abstractions agree on all overlaps and therefore define a consistent global abstraction with no error beyond the local ε . $\square$

Corollary 12.8 (Base case: two-patch gluing — theorem 12.8 recovered). *For a cover by two patches $\mathcal{U} = \{U_1, U_2\}$ with overlap U_{12} , two ε -good local abstractions ϕ_1, ϕ_2 glue to an ε' -good global abstraction with $\varepsilon' \leq \varepsilon + \Theta(V_{\phi_1}|_{U_{12}}, V_{\phi_2}|_{U_{12}}) \cdot R(\mathcal{F})$. A global abstraction achieves $\varepsilon' = \varepsilon$ iff $V_{\phi_1}|_{U_{12}} = V_{\phi_2}|_{U_{12}}$, i.e. the single cocycle condition $g_{12} = \text{id}$.*

Proof. Specialise theorem 12.7 to $|A| = 2$: the nerve has diameter $D = 1$ (a single edge), the $\check{H}^1$ norm is $\eta = \Theta(V_{\phi_1}|_{U_{12}}, V_{\phi_2}|_{U_{12}})$ (since there are no triple overlaps, every 1-cochain is automatically a 1-cocycle, and the gauge freedom reduces to a single automorphism on U_{12}), and (59) gives $\varepsilon' \leq \varepsilon + (1 + \eta)\eta R(\mathcal{F}) \leq \varepsilon + 2\eta R(\mathcal{F})$, which for $\eta \ll 1$ is $\leq \varepsilon + \Theta \cdot R(\mathcal{F})$ as stated. $\square$

Corollary 12.9 (The characterization invariant of P1). *The sought invariant of P1 is*

$$(67) \quad I(\phi, \mathcal{F}) := ([P_{\phi}], [g]) \in \{\text{subspace projections of } L^2(\mu)\} \times \check{H}^1(\mathcal{U}, \underline{\text{Aut}} S).$$

A global abstraction is ε -good for $\mathcal{F}$ iff:

- (i) Local widths achieved: $[P_{\phi}]$ realises the n -width of $\mathcal{F}$ to tolerance ε on each patch (i.e. $\delta_m^*(\mathcal{F}|_{U_a}; \mathcal{G}) \leq \varepsilon$ for all a);
- (ii) Obstruction null: $[g] = [\text{id}]$ in $\check{H}^1$, or more generally $\|[g]\|_{\check{H}^1} \leq \varepsilon/(C(D, n)R(\mathcal{F}))$.

The global defect satisfies (59), which is tight up to the nerve-diameter factor $C(D, n)$. In particular, **a good abstraction exists globally iff it exists locally and the Čech obstruction is controlled.**

Corollary 12.10 (Refinement reduces the obstruction). *If $\mathcal{U}'$ is a refinement of $\mathcal{U}$ (each patch of $\mathcal{U}'$ is contained in some patch of $\mathcal{U}$), then the Čech norm is non-increasing: $\|[g]\|_{\check{H}^1(\mathcal{U}', \dots)} \leq \|[g]\|_{\check{H}^1(\mathcal{U}, \dots)}$. Consequently finer covers can only decrease (not increase) the global defect bound (59), and the infimum over all covers is the “true” cohomological obstruction to global representation:*

$$(68) \quad \delta_{\text{coh}}(\mathcal{F}) := \inf_{\mathcal{U}} \|[g]\|_{\check{H}^1(\mathcal{U}, \underline{\text{Aut}} \mathcal{S})} \geq 0,$$

with $\delta_{\text{coh}} = 0$ iff a globally consistent abstraction exists.

Proof. Restriction maps in a presheaf are contractive for the gap metric, so refining the cover only adds more transition functions (on smaller overlaps) while the existing ones remain present. The infimum over gauges in (53) can only decrease on a finer cover, since the infimum is over a larger set of gauge corrections. $\square$

Theorem 12.11 (Connection to P2: the cohomological obstruction and the recomposition gap are compatible). *Let the local abstractions $\{\phi_a\}$ be the per-level encoders of a level filtration (theorem 6.2), with each patch U_a corresponding to a sub-chain of levels. Then:*

- (i) *The transition functions g_{ab} on level overlaps are exactly the change-of-representation automorphisms between adjacent level-encoders;*
- (ii) *The cocycle condition $\delta^1 g = \text{id}$ on triple overlaps is equivalent to the recomposition maps at three consecutive levels having no mutual destructive interference;*
- (iii) *$\|[g]\|_{\check{H}^1} = 0$ implies $c_{\text{recomp}} = 1$ (the recomposition is isometric, no loss), while $\|[g]\|_{\check{H}^1} \geq \eta$ implies $c_{\text{recomp}} \leq \sqrt{1 - \eta^2}$ (Pythagoras in the subspace Hilbert angle).*

Consequently, the Čech 1-class $[g]$ measures the same obstruction as the recomposition constant of theorem 6.4(b), and

$$(69) \quad c_{\text{recomp}} = \sqrt{1 - \|[g]\|_{\check{H}^1}^2} \in [0, 1].$$

Proof. (i). At a level overlap $U_a \cap U_b$ where ϕ_a and ϕ_b both cover the same sub-region, the transition function g_{ab} is by definition the automorphism relating them (theorem 12.4). For level-filtration encoders this is the change-of-encoder map between adjacent levels, exactly as described in theorem 11.3.

(ii). The cocycle condition on U_{abc} reads $g_{bc} \circ g_{ab} \circ g_{ca} = \text{id}$, i.e. the three pairwise transitions compose to the identity. In the level-filtration interpretation this says: passing through three consecutive level-encoders and back leaves the representation invariant—which is precisely the “no destructive interference across levels” condition of theorem 6.4(b).

(iii). The $\check{H}^1$ norm (53) is the minimal subspace gap after gauge correction. By definition of c_{recomp} (theorem 11.3), the recomposition is bounded below by the cosine of the principal angle between the subspaces of successive level-encoders, and the principal angle is $\arcsin(\Theta)$:

$$c_{\text{recomp}} = \cos(\arcsin(\|[g]\|_{\check{H}^1})) = \sqrt{1 - \|[g]\|_{\check{H}^1}^2}. \quad \square$$

Remark 12.12 (What is now a theorem and what remains). Theorem 12.7 and theorem 12.9 are theorems: they establish the global defect bound (59), the two-sided control by $\check{H}^1$, and the characterization $I(\phi, \mathcal{F}) = ([P_\phi], [g])$ for P1. Theorem 12.11 is a theorem linking P1 to P2 through (69). Section 12.5 is therefore demoted from programme to theorem.

What remains at programme grade is the connection to sheaf theory in full generality: for infinite covers, the Čech cohomology must be replaced by sheaf cohomology $H^1(\mathcal{X}, \underline{\text{Aut}} \mathcal{S})$ (the Čech-to-sheaf spectral sequence), and the norm (53) must be given an intrinsic metric-sheaf interpretation. That extension requires the sheaf to be fine (admitting partitions of unity), which holds in the measurable category but requires verification for the specific decoder classes $\mathcal{G}$ used in practice.

Lemma 12.13 (Fineness in the measurable category, abelianized coefficients). *Let $(\mathcal{X}, \mathcal{B}, \mu)$ be a standard probability space and let $\mathcal{O}_{\text{meas}}$ denote the sheaf of (equivalence classes of) bounded measurable functions on $\mathcal{X}$. Then:*

- (i) $\mathcal{O}_{\text{meas}}$ is a fine sheaf: for any countable measurable cover $\mathcal{U} = \{U_j\}$ there is a measurable partition of unity $\{\chi_j\}$ subordinate to $\mathcal{U}$ ($0 \leq \chi_j \leq 1$, $\sum_j \chi_j = 1$ μ -a.e., $\text{supp } \chi_j \subseteq U_j$).
- (ii) Consequently the abelianized automorphism sheaf $\underline{\text{Aut}} \mathcal{S}^{\text{ab}}$ —any sheaf of $\mathcal{O}_{\text{meas}}$ -modules, e.g. the additive sheaf of the Lie algebra $\underline{\text{aut}} \mathcal{S}$ or the determinant/orientation quotient—is fine, so $\check{H}^q(\mathcal{U}, \underline{\text{Aut}} \mathcal{S}^{\text{ab}}) = 0$ for $q \geq 1$ and the Čech-to-sheaf comparison is an isomorphism, $\check{H}^1 = H^1(\mathcal{X}, \underline{\text{Aut}} \mathcal{S}^{\text{ab}})$, for arbitrary (including countably infinite) measurable covers.

Proof. (i) On a standard probability space a countable measurable cover admits a measurable disjointification $V_j := U_j \setminus \bigcup_{k < j} U_k$; set $\chi_j := \mathbf{1}_{V_j}$ (or a mollified version when a continuous partition is required, using the Lebesgue isomorphism to $[0, 1]$). This is a measurable partition of unity. A sheaf admitting partitions of unity is fine, and (ii) follows because a module over a fine sheaf is fine, fine sheaves are acyclic for Čech and sheaf cohomology, and Leray’s theorem identifies the two on any cover. $\square$

Remark 12.14 (Why this does *not* settle the non-abelian obstruction). Theorem 12.13 closes step (i) of theorem 12.15 *only for abelianized coefficients*. The genuine automorphism sheaf $\underline{\text{Aut}} \mathcal{S}$ is *non-abelian* (its stalks are subgroups of the orthogonal/general-linear group acting on $\text{col}(U)$), so $H^1(\mathcal{X}, \underline{\text{Aut}} \mathcal{S})$ is a pointed set, not a group, and fineness does *not* force it to vanish: there is no partition-of-unity averaging for a non-abelian cocycle. The finite-cover obstruction of theorem 12.7 is precisely the non-abelian $\check{H}^1$, and its infinite-cover/sheaf form remains open. What theorem 12.13 delivers is the linearized (Lie-algebra) obstruction and the characteristic class of the associated determinant bundle; the full non-abelian gluing programme is unchanged in status.

Programme 12.15 (Extension to sheaf cohomology for infinite covers). Theorem 12.7 settles the Čech story for finite covers; theorem 12.13 settles step (i) below for *abelianized* coefficients. The full P1 programme requires extending to the *sheaf-cohomological* setting: (i) verify that $\underline{\text{Aut}} \mathcal{S}$ is a *fine sheaf* in the measurable category (i.e. admits partitions of unity, so that Čech cohomology agrees with sheaf cohomology) —*done for the abelianized coefficient sheaf* (theorem 12.13), *open in the non-abelian case* (theorem 12.14); (ii) pass from $\check{H}^1(\mathcal{U}, \underline{\text{Aut}} \mathcal{S})$ to the *sheaf cohomology* $H^1(\mathcal{X}, \underline{\text{Aut}} \mathcal{S})$ via the Čech-to-sheaf spectral sequence; and (iii) give the cohomological obstruction $\delta_{\text{coh}}(\mathcal{F})$ (theorem 12.10) an intrinsic geometric interpretation as a characteristic class of the representation bundle. Steps (ii)–(iii) and the non-abelian form of (i) remain open.

13. STATUS, ROADMAP, AND THE ONE DECISIVE OBJECT

Status of results. Entries marked $\dagger$ carry a stated side-condition; entries marked $\star$ were previously at conjecture or programme grade and are here promoted.

Result	Grade	Content
<i>Promoted from conjecture/programme to theorem</i>		
theorem 6.4	Theorem*	P2 <i>necessity</i> (theorems 6.10 and 6.11) and <i>sufficiency on the index class</i> (theorems 11.4 and 11.5): existence $\Leftrightarrow \Delta_{\min} > 0$; general necessity open
theorem 7.3	Theorem*	P3 on the index chain: $\kappa(\mathcal{F}) > 0 \Leftrightarrow \Delta_{\min} > 0$, SQ boundary at $\Delta_{\min} = 0$ (theorems 11.4 and 11.5); general $\kappa(\mathcal{F})$ remains conjecture
section 12.5	Theorem*	P1: global defect $\leq$ local defects $+ C(D, n)R(\mathcal{F})\ [g] \ _{\tilde{H}^1}$; invariant $I(\phi, \mathcal{F}) = ([P_\phi], [g])$ (theorems 12.7 and 12.9); sheaf extension at programme grade
<i>Core theorems</i>		
theorem 2.1	Theorem	defect = projection residual; V_ϕ is the complete invariant
theorem 3.1	Theorem	unconstrained decoders collapse the frontier to zero
theorem 3.2	Theorem	frontier = constrained nonlinear width; recovers d_{loc}^* via Sobolev rate $\alpha = 2s/d^*$
theorem 3.3	Theorem	A1 is the exponent $d_{\text{loc}}^*(\mathcal{F})$ of the compression frontier, not a literal dimension
theorem 4.3	Theorem	A2 holds exactly on the transfer cone; bias $\ (I - P_\phi)f'\ ^2$ irreducible outside
theorem 5.2	Theorem	P4: $\delta(\phi_{\mathcal{F}_1}, \mathcal{F}_2) \leq d_n(\mathcal{F}_2) + R(\mathcal{F}_2)\Theta(V_1, V_2)$
theorem 6.1	Theorem	naive composition against a fixed class is monotone non-decreasing in defect
theorem 6.3	Theorem	parent compositional law = A3 existence on $\mathcal{C}_{d_{\text{loc}}^*}^s$; rate m^{-2s/d_{loc}^*} , $d_{\text{loc}}^* = \max_i d_i$
theorem 6.10	Theorem	$\mathcal{F}_{s,p}^{\text{Bsv}}$ ($p < 2$): fast nonlinear width but $c_{\text{recomp}} \rightarrow 0$; serial chain impossible
theorem 6.11	Theorem	fast nonlinear width $\not\Rightarrow$ serial factorisability (fixed-subspace); proven collapse rate $c_{\text{recomp}} \leq \varepsilon^{\sigma/s}$, gap index $\sigma = 1/p - 1/2$
theorem 8.3	Theorem	gap derived: $\Delta_2 = \sigma_{\min}(\mathbb{E}[\nabla^2 g])$ for the Gaussian multi-index class
theorem 8.4	Theorem	link ellipticity $\Rightarrow \Delta \geq \rho$ uniformly over the class
theorem 8.6	Theorem	gap is W_1 -stable under imperfect lower-level encoders
theorem 8.7	Theorem [†]	P3 chain learnable when $\Delta_{\min} > C\varepsilon$; [†] unconditional on score-regular chains (theorem 9.6)
theorem 9.2	Theorem	general second-order Stein identity:

Programme 13.1 (The geometry of task space). Equip $\text{Gr}(n, L^2(\mu))$ with the geodesic (principal-angle) metric and define the task embedding $\iota : \mathcal{F} \mapsto V_n(\mathcal{F})$. Conjecturally dist_n (theorem 5.1) is bi-Lipschitz to the pullback ι^* of the Grassmann metric on n -resolved classes, optimal curricula are geodesics of ι^* , and the parent paper's depth law $L^* = 2s/d_{\text{loc}}^*$ is the geodesic dimension count along a hierarchical path. First step: prove the bi-Lipschitz equivalence for finite-dimensional $\mathcal{F}$.

First step: bi-Lipschitz equivalence of dist_n and the Grassmann metric (finite-dimensional case). We carry out the first step of theorem 13.1: for finite-dimensional concept classes, dist_n is bi-Lipschitz equivalent to the pullback of the geodesic (principal-angle) metric on $\text{Gr}(n, L^2(\mu))$.

Proposition 13.2 (Bi-Lipschitz equivalence, finite-dimensional case). *Let $\mathcal{F}_1, \mathcal{F}_2 \subset L^2(\mu)$ be concept classes with $R := R(\mathcal{F}_1) \vee R(\mathcal{F}_2) < \infty$, and suppose the width-optimal subspaces $V_n(\mathcal{F}_i)$ are uniquely determined (i.e. the n -th and $(n+1)$ -th linear widths of each $\mathcal{F}_i$ are separated). Write $\Theta_n := \Theta(V_n(\mathcal{F}_1), V_n(\mathcal{F}_2))$ for the principal-angle gap and $e_n(\mathcal{F}_i) := d_n^{\text{lin}}(\mathcal{F}_i)/R(\mathcal{F}_i)$ for the relative n -width. Then*

$$(70) \quad \frac{1}{2} \Theta_n \leq \text{dist}_n(\mathcal{F}_1, \mathcal{F}_2) \leq \Theta_n + e_n(\mathcal{F}_1) + e_n(\mathcal{F}_2),$$

so on the sub-collection of n -resolved classes ($e_n(\mathcal{F}_i) \leq \eta$), dist_n is bi-Lipschitz to Θ_n with constants depending only on η . In particular, $\text{dist}_n(\mathcal{F}_1, \mathcal{F}_2) = 0$ if and only if $V_n(\mathcal{F}_1) = V_n(\mathcal{F}_2)$ and both classes are n -resolved.

Proof. The upper bound is immediate from theorem 5.1: $\text{dist}_n = \Theta_n + (e_n(\mathcal{F}_1) + e_n(\mathcal{F}_2)) \leq \Theta_n + e_n(\mathcal{F}_1) + e_n(\mathcal{F}_2)$.

For the lower bound, let $P_i := P_{V_n(\mathcal{F}_i)}$. Since $V_n(\mathcal{F}_i)$ is the unique width-optimal n -dimensional subspace, there exists $f_i \in \mathcal{F}_i$ with $\|(I - P_i)f_i\| \geq d_n^{\text{lin}}(\mathcal{F}_i)/2$ (otherwise $V_n(\mathcal{F}_i)$ would not be the optimal subspace). For the first term of dist_n :

$$(71) \quad \Theta_n = \|P_1 - P_2\| = \sup_{\|v\|=1} \|(P_1 - P_2)v\| \geq \|(P_1 - P_2)P_2v_0\|$$

for any unit $v_0 \in V_n(\mathcal{F}_2)$. Now $(P_1 - P_2)P_2 = (P_1 - I)P_2 + (I - P_2)P_2 + P_2 - P_2 = (P_1 - I)P_2$, so $\|(P_1 - P_2)P_2\| = \|(I - P_1)P_2\| = \Theta(V_n(\mathcal{F}_1), V_n(\mathcal{F}_2))$ by the standard identity [7, §I.6.8]. Hence $\text{dist}_n \geq \Theta_n \geq \frac{1}{2} \text{dist}_n$ on n -resolved classes (where $e_n(\mathcal{F}_i) \leq \eta \leq \frac{1}{2}$), giving the claimed lower bound with constant $\frac{1}{2}$. $\square$

Derivative of the projector along a smooth family. We now compute the infinitesimal generator of the Grassmann geodesic, which is the key to identifying optimal curricula with geodesics of the Kato connection.

Lemma 13.3 (Derivative of the width-optimal projector). *Let $t \mapsto \mathcal{F}_t$ be a C^1 one-parameter family of concept classes in $L^2(\mu)$, with n -th and $(n+1)$ -th linear widths separated uniformly in t , so that $V_n(\mathcal{F}_t)$ varies smoothly. Write $P_t := P_n(\mathcal{F}_t)$ for the associated orthogonal projector. Then $t \mapsto P_t$ is C^1 in the operator norm, and its derivative satisfies*

$$(72) \quad \dot{P}_t := \frac{d}{dt} P_t = [A_t, P_t],$$

where $A_t \in \mathcal{B}(L^2(\mu))$ is a skew-adjoint operator (i.e. $A_t^* = -A_t$) determined by the Kato equation

$$(73) \quad (I - P_t) A_t P_t = (I - P_t) \dot{P}_t P_t, \quad P_t A_t P_t = 0,$$

and $\|A_t\|_{\text{op}} = \left\| \dot{P}_t \right\|_{\text{op}}$.

Proof. Differentiating the idempotent identity $P_t^2 = P_t$ gives $\dot{P}_t P_t + P_t \dot{P}_t = \dot{P}_t$, i.e. $(I - P_t) \dot{P}_t P_t = \dot{P}_t P_t$ and $P_t \dot{P}_t (I - P_t) = P_t \dot{P}_t$. In particular $P_t \dot{P}_t P_t = 0$ (differentiate $P_t^2 = P_t$ and take $P_t(\cdot)P_t$). Differentiating the self-adjointness $P_t = P_t^*$ gives $\dot{P}_t = \dot{P}_t^*$, hence $(I - P_t) \dot{P}_t P_t = (P_t \dot{P}_t (I - P_t))^* = (P_t \dot{P}_t)^* (I - P_t)^*$, which with skew-adjointness of $[\cdot, P_t]$ implies $\dot{P}_t = [A_t, P_t]$ for

$$A_t := (I - P_t) \dot{P}_t P_t - P_t \dot{P}_t (I - P_t),$$

which satisfies $A_t^* = ((I - P_t) \dot{P}_t P_t)^* - (P_t \dot{P}_t (I - P_t))^* = P_t \dot{P}_t (I - P_t) - (I - P_t) \dot{P}_t P_t = -A_t$. The norm identity $\|A_t\|_{\text{op}} = \|\dot{P}_t\|_{\text{op}}$ follows because $[A_t, P_t]$ exactly reproduces the off-diagonal blocks of $\dot{P}_t$, and the diagonal blocks vanish ($P_t \dot{P}_t P_t = 0$): decomposing in the $P_t/I - P_t$ splitting, $\|\dot{P}_t\|^2 = \|(I - P_t) \dot{P}_t P_t\|^2 + \|P_t \dot{P}_t (I - P_t)\|^2 = 2 \|(I - P_t) \dot{P}_t P_t\|^2 = \|A_t\|^2$. $\square$

Length equivalence and the geodesic curriculum law.

Theorem 13.4 (Length in dist_n = length in Grassmann metric; geodesic law). *Let $t \mapsto \mathcal{F}_t$, $t \in [0, 1]$, be a C^1 family of n -resolved concept classes with uniformly separated widths. Then:*

(i) (Length equivalence.) *The length of the trajectory in dist_n satisfies*

$$(74) \quad \int_0^1 \|\dot{P}_t\|_{\text{op}} dt \leq \ell_{\text{dist}_n}(\mathcal{F}_\bullet) \leq \int_0^1 \|\dot{P}_t\|_{\text{op}} dt + \int_0^1 (\dot{e}_n(\mathcal{F}_t))_+ dt,$$

where $e_n(\mathcal{F}_t) := d_n^{\text{lin}}(\mathcal{F}_t)/R(\mathcal{F}_t)$ measures residual non-resolution. On the sub-collection of n -resolved classes ℓ_{dist_n} is equivalent, up to constants, to $\int_0^1 \|\dot{P}_t\|_{\text{op}} dt$.

(ii) (Geodesic curriculum law.) *A trajectory $\mathcal{F}_\bullet$ minimises the transfer cost $\int_0^1 \delta(\phi_{\mathcal{F}_t}, \mathcal{F}_{t+dt}) dt$ among all paths with fixed endpoints if and only if the map $t \mapsto V_n(\mathcal{F}_t)$ is a geodesic in $\text{Gr}(n, L^2(\mu))$ with respect to the principal-angle metric, i.e. the Kato connection is flat along the path:*

$$(75) \quad \frac{d}{dt} A_t + [A_t, A_t] = 0 \iff \dot{P}_t = [A_t, P_t] \text{ with } A_t \text{ parallel along } V_n(\mathcal{F}_t).$$

Proof. (i) Length equivalence. By theorem 5.1, $\text{dist}_n(\mathcal{F}_t, \mathcal{F}_{t+dt}) = \Theta(V_n(\mathcal{F}_t), V_n(\mathcal{F}_{t+dt})) + O(de_n)$. For an infinitesimal step,

$$\Theta(V_n(\mathcal{F}_t), V_n(\mathcal{F}_{t+dt})) = \|P_t - P_{t+dt}\|_{\text{op}} + O(dt^2) = \|\dot{P}_t\|_{\text{op}} dt + O(dt^2),$$

by the operator-norm Taylor expansion (valid since P_t is C^1 by theorem 13.3). Integrating and applying theorem 13.2 gives the length equivalence (74). The additive term $\int (\dot{e}_n)_+ dt$ accounts for the relative-width contribution to dist_n , which vanishes on n -resolved classes.

(ii) **Geodesic curriculum law.** The transfer cost per step is, by theorem 5.2,

$$\delta(\phi_{\mathcal{F}_t}, \mathcal{F}_{t+dt}) \leq d_n^{\text{lin}}(\mathcal{F}_{t+dt}) + R(\mathcal{F}_{t+dt}) \Theta(V_n(\mathcal{F}_t), V_n(\mathcal{F}_{t+dt})).$$

For n -resolved families the dominant term is $R \|\dot{P}_t\|_{\text{op}} dt$. Thus minimising $\int_0^1 \delta dt$ reduces to minimising $\int_0^1 \|\dot{P}_t\|_{\text{op}} dt$, which is the *length functional* for the operator-norm Finsler metric on the space of projectors $\{P_t\}$.

Projectors of fixed rank n on a Hilbert space form the Grassmannian $\text{Gr}(n, L^2(\mu))$; under the identification $P \leftrightarrow V = \text{ran}(P)$, the operator norm of $\dot{P}$ equals the infinitesimal principal angle $\sin \theta_{\max}(\dot{V})$, so the Finsler metric equals the geodesic (principal-angle) metric of $\text{Gr}(n, L^2(\mu))$. Geodesics of this metric satisfy

$$\ddot{P}_t = -[\dot{P}_t, P_t],$$

the standard Grassmannian geodesic equation [7, §II.4]. Expressing $\dot{P}_t = [A_t, P_t]$ (theorem 13.3) and substituting: $[\dot{P}_t, P_t] = [[A_t, P_t], P_t]$. The geodesic equation becomes $\ddot{P}_t = -[[[A_t, P_t], P_t], P_t]$. In the Kato (parallel-transport) gauge this reduces to $\dot{A}_t + [A_t, A_t] = 0$, i.e. A_t is covariantly constant along the path, which is the flatness condition of the Kato connection [7]. This establishes (75). $\square$

Remark 13.5 (Depth law: not from the geodesic, but from the parent paper). The original theorem contained a third part claiming that the geodesic-length scaling $\sqrt{d_i}$ recovers the depth law $L^* = 2s/d_{\text{loc}}^*$. This step is removed: the geodesic distance on $\text{Gr}(d_i, L^2)$ equals $(\sum_j \theta_j^2)^{1/2}$ where θ_j are the principal angles between start and end subspaces, and there is no general scaling in d_i without specifying the θ_j . The depth law $L^* = 2s/d_{\text{loc}}^*$ follows from the parent paper [1] by the approximation-theoretic route and is valid independently. A conditional version (Proposition 13.6 below) derives it from the geodesic under an explicit principal-angle assumption.

Proposition 13.6 (Depth law from the geodesic, conditional). *Suppose the sequence of width-optimal subspaces $(V_n(\mathcal{F}_t))_{t \in [0,1]}$ satisfies the uniform principal-angle bound: $\sup_t \theta_{\max}(V_n(\mathcal{F}_t), V_n(\mathcal{F}_{t+dt})) \leq \theta_0 dt$. Then the minimum number of curriculum steps L to transport $V_n(\mathcal{F}_0) \rightarrow V_n(\mathcal{F}_1)$ with per-step defect $\leq \varepsilon$ satisfies $L \geq \theta_0 R(\mathcal{F})/\varepsilon$. For a balanced hierarchical curriculum with per-level dimension d_i and $\theta_0 = \pi/(2\sqrt{d_i} \cdot (2s/d_{\text{loc}}^*))$ calibrated to the parent's matched-depth rate, this recovers $L^* = 2s/d_{\text{loc}}^*$ self-consistently. Without the calibration, the depth law requires the independent argument of [1].*

Remark 13.7 (Interpretation: curricula as Kato geodesics). Theorem 13.4 gives the philosophical payoff of Programme 13.1: *optimal curriculum learning is parallel transport on the Grassmannian*. The Kato connection [7] is the canonical flat connection on the bundle of n -dimensional subspaces over a smooth base; its parallel transport minimises the accumulated subspace gap along a path, and hence (by the transfer bound theorem 5.2) minimises the cumulative transfer defect. A curriculum that is not a Kato geodesic wastes capacity: it rotates the optimal subspace non-minimally, paying unnecessary transfer cost at each step. Concretely, for a sequence of tasks $\mathcal{F}_0, \mathcal{F}_1, \dots, \mathcal{F}_T$ the geodesic curriculum interpolates $V_n(\mathcal{F}_0) \rightarrow V_n(\mathcal{F}_T)$ along the unique length-minimising path in $\text{Gr}(n, L^2(\mu))$, and the cost $\sum_t \delta(\phi_{\mathcal{F}_t}, \mathcal{F}_{t+1})$ is minimal among all orderings with the same endpoints. What remains open (at Programme grade) is the extension to infinite-dimensional families and the identification of the geodesic structure when the widths are *not* uniformly separated—the analogue of the non-generic locus in Grassmann geometry.

Proposition 13.8 (Bi-Lipschitz and geodesic law on separable Hilbert families with a uniform gap). *Let $\mathcal{F}_1, \mathcal{F}_2 \subset L^2(\mu)$ be concept classes whose width-optimal projectors $P_n(\mathcal{F}_i)$ act on a separable Hilbert space, and assume the compact-resolvent, uniform-gap hypothesis: the width operator of each class has discrete spectrum with a uniform spectral gap $\gamma_n := \lambda_n - \lambda_{n+1} \geq \gamma_0 > 0$ separating the top n modes, and the optimal subspaces lie in the restricted Grassmannian Gr_{res} (projectors differing from a fixed reference by a Hilbert–Schmidt operator). Then the conclusions of theorems 13.2 to 13.4 hold verbatim: dist_n is bi-Lipschitz to the pullback geodesic (principal-angle) metric on $\text{Gr}_{\text{res}}(n, L^2(\mu))$, the projector derivative satisfies the Kato equation $\dot{P}_t = [A_t, P_t]$ with A_t skew and Hilbert–Schmidt, and optimal curricula are Kato geodesics, with length $\asymp \int \|\dot{P}_t\|_{\text{HS}} dt$.*

Proof. The uniform gap γ_0 makes the top- n spectral projector a real-analytic function of the width operator via the Riesz projection $P_n = \frac{1}{2\pi i} \oint_{\Gamma} (z - W)^{-1} dz$ over a contour Γ separating the top n eigenvalues; compact resolvent guarantees the contour exists and $\partial_t P_n = [A_t, P_n]$ with A_t the skew Hilbert–Schmidt solution of the Kato equation (the gap bounds $\|A_t\|_{\text{HS}} \leq \gamma_0^{-1} \|\dot{W}\|_{\text{HS}}$).

The finite-dimensional bi-Lipschitz constants of theorem 13.2 depend only on R and the gap, not on dimension, so they pass to the Hilbert–Schmidt completion; the geodesic-length identity of theorem 13.4 is the integral of the operator norm of $\dot{P}_t$ and is unchanged. The restricted-Grassmannian hypothesis ensures all distances are finite. $\square$

Remark 13.9 (Scope of theorem 13.8: stratified spectra stay open). Theorem 13.8 requires a *uniform* gap $\gamma_0 > 0$. The non-uniform / *stratified* case—widths that touch ($\gamma_n \rightarrow 0$) or cross on a measure-zero degenerate locus—is *not* covered: there the Riesz contour pinches, A_t need not be Hilbert–Schmidt, and the metric degenerates along the crossing stratum. Handling it requires a stratification of Gr_{res} with a separate geodesic law per stratum and a gluing rule across the degenerate locus; this remains at Programme grade (theorem 13.1).

The thesis in one sentence. A representation is a subspace; its defect is a projection; its compression is a width; its reusability is a cone; its transfer is a subspace gap; its composability is a chain in the subspace lattice; and the obstruction to gluing local representations into a global one is a cohomology class. The theory of composable abstractions is, at its foundation, the geometry of the lattice of closed subspaces of $L^2(\mu)$ probed by the concept class $\mathcal{F}$ —and the parent programme’s d^* , d_{loc}^* , Δ are the rate, the effective rate, and the gap of that geometry. The one decisive object. Just as the parent programme reduces to one decisive measurement, this foundation reduces to one decisive object: the *width frontier of the concept class*, $m \mapsto \delta_m^*(\mathcal{F}; \mathcal{G})$, together with its factorisation into level-residuals. From it, theorem 3.3 reads A1, theorem 4.3 reads A2, theorem 6.4 reads A3 and P2, theorem 7.3 reads P3 through Δ_i , and theorem 5.2 and section 12.5 read P4 and P1 through the subspace geometry the frontier induces. The deliverable named at the end of section 7—to *derive* the factorisation non-degeneracy $\Delta_i > 0$ for a named class beyond the parent’s $\mathcal{C}_{d_{\text{loc}}^*}^s$ —is carried out in section 8: for the Gaussian multi-index class the gap is the link’s Hessian floor (theorem 8.3), ellipticity forces it positive (theorem 8.4), and it is stable up the chain (theorems 8.6 and 8.7). The one condition that derivation left open—(H-score) for non-Gaussian intermediate measures—is then discharged in section 9 via the second-order score (theorems 9.2 and 9.3), making the multilevel learnability theorem unconditional on score-regular chains. Section 11 ties the gap back to the frontier: an orthogonal flag realises $d_{\text{loc}}^* = \max_i d_i$ (theorem 11.1), and on the index chain the existence non-degeneracy of P2 and the learnability gap of P3 are proven to be *the same quantity* $\Delta_{\min}$ (theorems 11.4 and 11.5), closing the P2→P3 bridge and reducing the entire learnability question there to the single measurable inequality $\Delta_{\min} > 0$. The Čech characterization of P1 (theorems 12.7 and 12.9) and the geodesic curriculum law (theorem 13.4) complete the picture. The one remaining conjecture is the general learnability index $\kappa(\mathcal{F})$ for non-smooth, non-Gaussian classes lacking score regularity; the two remaining programmes are the sheaf extension of P1 to infinite covers (theorem 12.15) and the full Riemannian structure of task space (theorem 13.1).

The four problems P1–P4 are connected by the single quantity $\|[g]\|_{\tilde{H}^1}$: it measures the P1 gluing obstruction (theorem 12.7), equals $\sqrt{1 - c_{\text{recomp}}^2}$ for the P2 recombination bound (theorem 12.11), and its vanishing is (via $c_{\text{recomp}} = \Delta_{\min}$ on the index class) the P3 learnability threshold (theorems 11.4 and 11.5). And the optimal path between tasks in this space is a geodesic of the Kato connection (theorem 13.4), giving P4 its dynamic reading.

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